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IEEE Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors

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IEEE Power and Energy Society

Developed by the
Transmission and Distribution Committee

IEEE Std 738™-2023
(Revision of IEEE Std 738-2012)

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IEEE Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors

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Approved 29 June 2023

IEEE SA Standards Board

Abstract: A numerical method by which the core and surface temperatures of a bare overhead conductor are related to the steady or time-varying electrical current and weather conditions is described in this standard. Contents discussed also include how this method may be used to determine the conductor's current for a corresponding conductor temperature limit. Please note the content in this standard does not recommend suitable weather conditions or conductor parameters for use in line rating calculations.

Keywords: ampacity, bare overhead lines, current-temperature relationship, IEEE 738™, line ratings

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Introduction

This introduction is not part of IEEE Std 738-2023, IEEE Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors.

In 1986, IEEE Std 738™, IEEE Standard for Calculation of Bare Overhead Conductor Temperature and Ampacity Under Steady-State Conditions, was first published. The standard was developed "so that a practical sound, and uniform method (of calculation) might be utilized and referenced."

As part of the revision in 1993, the Working Group on the Calculation of Bare Overhead Conductor Temperatures, which was responsible for the revision of this standard, decided to address fault current and transient ratings and include their calculation in this standard. In the 2006 revision, SI units were added throughout, the solar heating calculation was extensively revised, and many editorial changes were made. Richard E. Kennon, James Larkey, Jerry Reding, and Dale Douglass did much of the revisions in the 1993 and 2006 versions of the standard.

The present version was modernized with the removal of the previously included "pseudo-code", updated sample calculations, revisions to improve consistency and clarity, and various other editorial updates.

Acknowledgments

Many persons have contributed to the preparation of this most recent standard. The primary contributors were Cody Davis, Paula Traynor, Alex Abboud, Justin Bell, Greg Bennett, Yair Berenstein, Joe Coffey, Nancy Fulk, Ravi Ganatra, Jake Gentle, Charles Holcombe, Drew Pearson, Jerry Reding, Jack Roughan, Ken Snider, and Paul Springer. Many other members of IEEE W.G. 15.11.02/06 "T&D Overhead Conductors & Accessories," chaired by Cody Davis and Nancy Fulk, contributed their time and thought. The 2012 revision was used as a resource by University of Toronto Electrical and Computer Engineering students, who provided numerous typographical corrections incorporated into this revision.

We would also like to recognize the contribution of the late Dale Douglass, who served as Chair of the working group for many years and was responsible for developing the 2012 revision of the standard.

Contents

IEEE Standard for Calculating the Current-Temperature Relationship of Bare Overhead Conductors

1. Overview

1.1 Scope

The standard describes a numerical method by which the core and surface temperatures of a bare overhead conductor are related to the steady or time-varying electrical current and weather conditions. The method may also be used to determine the conductor current which corresponds to conductor temperature limits. The standard does not recommend suitable weather conditions or conductor parameters for use in line rating calculations.

1.2 Purpose

The purpose of this document is to provide a standard method for the calculation of the current-temperature relationship of overhead power line conductors for steady-state and transient cases. This method may be used to develop line ratings, including dynamic ratings.

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).[6,7]

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

[6]The use of the word *must* is deprecated and cannot be used when stating mandatory requirements; *must* is used only to describe unavoidable situations.
[7]The use of *will* is deprecated and cannot be used when stating mandatory requirements; *will* is only used in statements of fact.

2. Definitions, acronyms, and abbreviations

2.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[8]

conductor temperature: The temperature of a conductor, T_{avg}, is normally assumed to be isothermal (i.e., no axial or radial temperature variation). In those cases where the current density exceeds 0.5 A/mm^2 (1 A/kcmil), especially for those conductors with more than two layers of aluminum strands, the difference between the core and surface may be significant. Also, the axial variation along the line may be important. Finally, for transient calculations where the time period of interest is less than 1 min with non-homogeneous aluminum conductor steel reinforced (ACSR) conductors, the aluminum strands may reach a high temperature before the relatively non-conducting steel core.

effective (radial) thermal conductivity: Effective radial thermal conductivity characterizes the bare stranded conductor's heterogeneous structure (including aluminum strands, air gaps, oxide layers) as if it were a single, homogeneous conducting medium. The use of effective thermal conductivity in the thermal model simplifies the calculation process and avoids complex calculations on a microscopic level including the assessment of contact thermal resistances between strands, heat radiation and convection in air gaps locked between strands.

heat capacity (material): When the average temperature of a conductor material is increased by dT as a result of adding a quantity of heat dQ, the ratio, dQ/dT, is the heat capacity of the conductor.

maximum allowable conductor temperature: The maximum conductor temperature limit that is selected for line rating purposes in order to limit loss of conductor strength, and which limits sag in order to maintain electrical clearances along the lines.

Reynolds number: A dimensionless number equal to air velocity times air density times conductor diameter divided by the kinematic viscosity of air, all expressed in consistent units. The Reynolds number, in this case, is equal to the ratio of inertia forces to the viscous force on the conductor. It is typically used to differentiate between laminar and turbulent flow.

specific heat: The specific heat of a conductor material is its heat capacity divided by its mass.

steady-state thermal rating: The constant electrical current that yields the maximum allowable conductor temperature for specified weather conditions and conductor characteristics under the assumption that the conductor is in thermal equilibrium (steady state).

thermal time constant: In response to a sudden change in current (or weather conditions), the conductor temperature will change in an approximately exponential manner, eventually reaching a new steady-state temperature if there is no further change. The thermal time constant is the time required for the conductor temperature to accomplish 63.2% this change. The exact change in temperature is not exponential so the thermal time constant is not used in the calculation described in this standard. It is, however, a useful concept in understanding line ratings.

time-varying weather and current: Neither weather conditions nor the electrical current carried by an overhead transmission line is typically constant over time. Yet both are assumed constant in conventional steady-state rating calculations. Even in the transient rating calculation where the current undergoes a step-change, the weather conditions are typically assumed constant. Only real-time rating methods consider the time-variation of line current and weather.

[8] *IEEE Standards Dictionary Online* is available at: http://dictionary.ieee.org. An IEEE Account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

transient thermal rating: The line current can change suddenly. The conductor temperature cannot. For short-time emergency line currents, the delay in heating the conductor may allow relatively high currents to be applied for short times (e.g., less than 3 thermal time constants) without exceeding the maximum allowable conductor temperature. The transient thermal rating is that emergency current (I_f) that yields the maximum allowable conductor surface or core temperature in a specified short time (typically less than 30 min) after a step change in electrical current from its initial current, I_i.

wind direction: Wind direction is relative to the conductor axis (where both wind direction and the conductor axis are assumed to be in a plane parallel to the earth). When the wind is blowing parallel to the conductor axis, it is termed "parallel wind." When the wind is blowing perpendicularly to the conductor axis, it is termed "perpendicular wind." In general, winds are neither parallel nor perpendicular to the line. At low wind speeds, where the wind is turbulent, it may have no persistent direction.

3. Units and identification of letter symbols

NOTE—SI units are the preferred unit of measure. However, in the United States the calculations described in this standard have frequently been performed using a combination of units that will be referred to throughout this document as "US" units. This standard includes both SI and US units with SI units preferred.[9]

Table 1—Units and identification of letter symbols

Symbol	Description	SI units	US units
A'	Projected area of conductor	m²/linear m	ft²/linear ft
C	Solar azimuth constant	deg	deg
C_{pi}	Specific heat of i^{th} conductor material	J/kg-°C	J/lb-°C
D_0	Outside diameter of conductor	m	ft
D_{core}	Conductor core diameter	m	ft
H_c	Solar altitude	deg	deg
H_e	Elevation of conductor above sea level	m	ft
I	Conductor current	A	A
I_f	Final current after step change	A	A
I_i	Initial current before step change	A	A
K_{angle}	Wind direction factor	—	—
K_{solar}	Solar elevation correction factor	—	—
k_f	Thermal conductivity of air at temperature T_{film}	W/(m-°C)	W/(ft-°C)
k_{th}	Effective radial thermal conductivity of conductor	W/(m-°C)	W/(ft-°C)
Lat	Degrees of latitude	deg	deg
mC_p	Total heat capacity of conductor	J/(m-°C)	J/(ft-°C)
m_i	Mass per unit length of i^{th} conductor material	kg/m	lb/ft
N	Day of the year	—	—
N_{Re}	Reynolds number	—	—
q_c	Convection heat loss rate per unit length	W/m	W/ft
q_{cn}	Natural convection heat loss rate per unit length	W/m	W/ft
q_{c1}	Low Reynolds number convection heat loss rate per unit length	W/m	W/ft
q_{c2}	High Reynolds number convection heat loss rate per unit length	W/m	W/ft

Table continues

[9]Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

Table 1—Units and identification of letter symbols *(continued)*

Symbol	Description	SI units	US units
q_r	Radiated heat loss rate per unit length	W/m	W/ft
q_s	Heat gain rate from sun	W/m	W/ft
Q_s	Total solar and sky radiated heat intensity at sea level	W/m^2	W/ft^2
Q_{se}	Total solar and sky radiated heat intensity corrected for elevation	W/m^2	W/ft^2
$R(T_{\text{avg}})$	Conductor resistance at temperature, T_{avg}	Ω/m	Ω/ft
T_a	Ambient air temperature	°C	°C
T_{avg}	Average temperature of aluminum strand layers	°C	°C
T_{core}	Conductor core temperature	°C	°C
T_f	Conductor temperature many time constants after step increase	°C	°C
T_{film}	Average temperature of the boundary layer	°C	°C
T_{high}	High average conductor temperature for which ac resistance is specified	°C	°C
T_i	Conductor temperature prior to step increase	°C	°C
T_{low}	Low average conductor temperature for which ac resistance is specified	°C	°C
T_s	Conductor surface temperature	°C	°C
V_w	Speed of air stream at conductor	m/s	ft/s
Y	Year	—	—
Z_c	Azimuth of sun	deg	deg
Z_l	Azimuth of line	deg	deg
Δt	Time step used in transient and dynamic calculations	s	s
ΔT_c	Conductor temperature increment corresponding to time step	°C	°C
α	Solar absorptivity	—	—
δ	Solar declination	deg	deg
ε	Emissivity	—	—
τ	Thermal time constant of the conductor	min	min
ϕ	Angle between wind and axis of conductor	deg	deg
β	Angle between wind and perpendicular to conductor axis	deg	deg
ρ_f	Density of air	kg/m^3	lb/ft^3
θ	Effective angle of incidence of the sun's rays	deg	deg
μ_f	Absolute (dynamic) viscosity of air[10]	kg/(m-s)	lb/(ft-s)
ω	Hour angle relative to solar noon	deg	deg
χ	Solar azimuth variable	—	—

4. Temperature and current calculation methods

The mathematical models of this standard are based upon the House and Tuttle [B26][11] method as modified by ECAR [B38]. To differentiate laminar and turbulent air flow, House and Tuttle uses two different equations for forced convection with a transition at a Reynolds number of 1000. This standard instead uses the modification

[10]Although the typical US units for dynamic viscosity of air are lb/(ft − hr), from this revision onwards this standard has adopted lb/(ft − s) for ease of calculating the dimensionless Reynolds number.
[11]Numbers in brackets correspond to those of the bibliography in Annex A.

in [B38] to make this transition where the two curves cross as shown in 4.4.3.1. More information on various calculation methods can be found in Annex B.

Conductor temperatures are a function of the following:

- Conductor material properties (primarily electrical conductivity)
- Conductor diameter
- Conductor surface condition (primarily emissivity and absorptivity)
- Weather conditions (air temperature, solar heating, wind speed and direction)
- Conductor electrical current

The conductor surface condition will change over time as the outermost layer of strands darken due to atmospheric pollution. Weather conditions vary greatly with the hour and season. Conductor electrical current varies with power system loading, generation dispatch, and other factors.

The standard acknowledges other factors that may affect the current-temperature relationship. This includes the calculation of electrical resistance, radial thermal gradients, diffuse and reflected solar heating, and precipitation cooling. A detailed discussion of these factors is provided in 4.5.

The equations relating electrical current to conductor temperature may be used to:

- Calculate the conductor temperature when the electrical current is known.
- Calculate the current (thermal rating) that yields a given maximum allowable conductor temperature.

The numerical thermal model presented in this standard is very general. It may be applied as follows:

- The "steady-state case" where the electrical current, conductor temperature, and weather conditions are assumed constant for all time.
- The "transient case" where the weather conditions are held constant but the electrical current undergoes a step change from an initial to a final value and the conductor temperature increases or decreases in a nearly exponential fashion from an initial temperature until it eventually reaches a new final temperature.
- The "dynamic case" where the conductor temperature is calculated for an electrical current and weather conditions which vary over time in any fashion.

As there is a great diversity of weather conditions and operating circumstances for which conductor temperatures and/or thermal ratings may be calculated, this standard does not undertake to list actual temperature-current relationships for specific conductors or weather conditions. Each user shall make their own assessment of which weather data and conductor characteristics best pertain to their area or particular transmission line. There are multiple methods of selection of these values as follows:

- Perform analysis of historical data.
- Use ISO required weather assumptions.
- *CIGRE Technical Brochure 299* [B8], provides guidance in selecting weather data for use in line rating calculations for maximum conductor temperatures up to 100 °C.
- Guidance from the conductor manufacturer with respect to the properties of the supplied conductor.

4.1 Steady-state case

4.1.1 Steady-state thermal rating

For a bare conductor, if the conductor's surface temperature (T_s), conductor characteristics, and the steady-state weather parameters (V_w, T_a, etc.) are given, the corresponding conductor current (I) to produce this conductor temperature can be calculated by the heat balance equation, Equation (1), shown in 4.4.1. The heat losses due to convection and radiation (q_c and q_r), the solar heat gain (q_s), and the conductor resistance R (T_{avg}) can be calculated by the equations of 4.4.3, 4.4.4, 4.4.5, and 4.4.6, respectively. While this calculation can be done for any conductor temperature and any weather conditions for which the heat transfer models are adequate, a maximum allowable conductor temperature (for which T_s and T_{avg} are set to when neglecting thermal gradients) and suitably conservative weather conditions are often used to calculate a steady-state thermal rating for the conductor.

4.1.2 Steady-state conductor temperature for a given current

Conductor resistance, convection, and thermal radiation are a function of conductor temperature (while solar heating is not). Therefore, if the conductor temperature is to be calculated for a given current rather than current being calculated for a given conductor temperature (as in 4.1.1), then, even if the conductor is assumed to be in steady-state, the heat balance equation shall be repeatedly solved for assumed conductor temperatures until the resulting current equals the given current as outlined in the steps below.

The use of a numerical solution method, as described in this standard, avoids the need to make complex and time-consuming approximations necessary to linearize the radiation and convection heat loss rates. For a given current, conductor characteristics, and steady-state weather conditions, the calculation process is straightforward:

a) The solar heat input to the conductor is calculated (as it is independent of conductor temperature) per 4.4.5.
b) A trial conductor temperature is assumed.
c) The conductor resistance is calculated for the trial temperature (see 4.4.6).
d) In combination with the assumed weather conditions, the convection and radiation heat loss terms are calculated per 4.4.3 and 4.4.4, respectively.
e) The resulting conductor current is calculated by means of the heat balance in Equation (1) of 4.4.1.
f) The calculated current is compared to the resulting conductor current.
g) The trial conductor temperature is then increased or decreased until the calculated current equals the specified current within a user-specified tolerance. This iteration is easily accomplished through the use of numerical iteration methods available in commercial spreadsheet software.

4.2 Transient case

4.2.1 Transient conductor temperature

The temperature of a bare overhead conductor is constantly changing in response to changes in electrical current and weather conditions (T_a, Q_s, V_w, ϕ). In the transient calculations described in this subclause, however, weather parameters are assumed to remain constant; and any change in electrical current is limited to a step change from an initial current, I_i, to a final current, I_f, at a time designated "$t = 0$" as illustrated in Figure 1.

Immediately prior to the current step change (t = 0−), the conductor is assumed to be in steady-state. Immediately after the current step change (t = 0+), the conductor temperature is unchanged (as are the conductor resistance and the heat loss rate due to convection and radiation), but the rate of heat generation

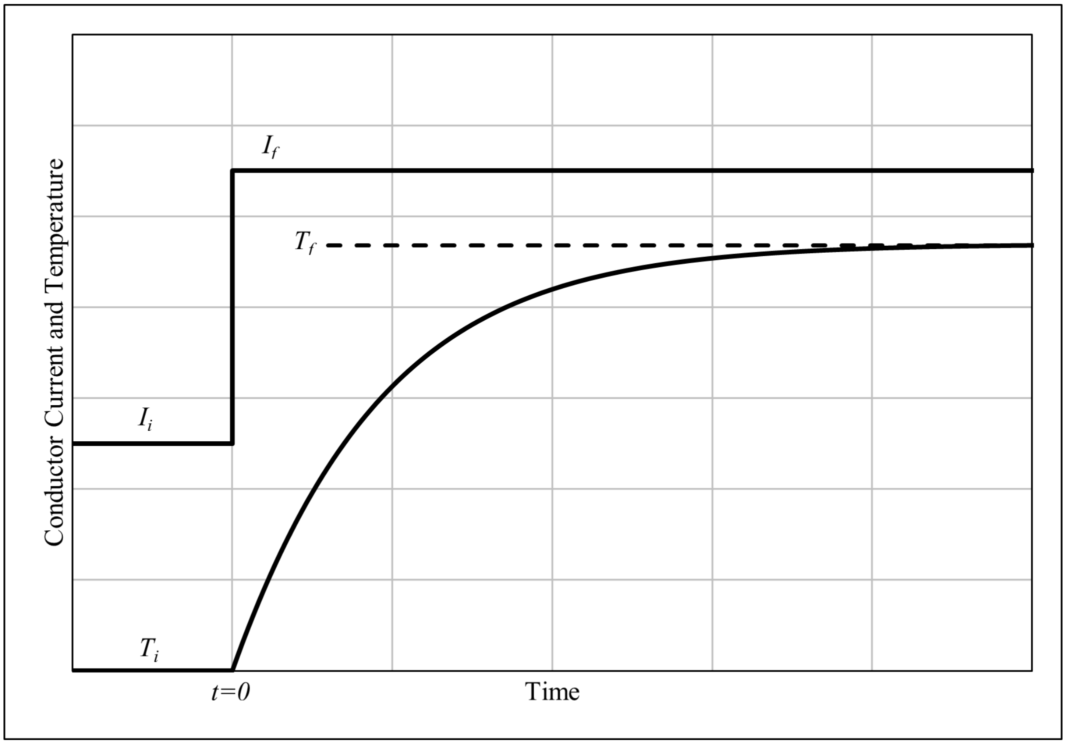

Figure 1—"Step" change in conductor current and corresponding change in conductor temperature

due to Joule heating, now $I_f^2 \times R(T_{avg})$, has increased. As a result of this thermal unbalance at time $t = 0+$, the heat, which cannot be immediately loss by convection or radiation, goes into conductor heat storage and the conductor temperature begins to increase at a rate given by the non-steady-state heat balance [Equation (3)] in 4.4.2.

After a period of time (time step), Δt, the conductor temperature has increased by a temperature change of ΔT_{avg}. The increased conductor temperature yields higher heat losses due to convection and radiation and somewhat higher Joule heat generation due to the increased conductor resistance. From Δt to $2\Delta t$, the conductor temperature continues to increase, but does so at a lower rate. After a large number of such time steps, the conductor temperature approaches its final steady-state temperature (T_f).

To calculate transient conductor temperature, the following steps are applied:

a) All heat terms [q_c, q_r, q_s, and $R(T_{avg})$] are calculated per the equations of 4.4 at the initial temperature T_i.

b) For the first time step in the increase in temperature ΔT_{avg}, Δt is calculated with these heat terms and the new current level, I_f. This temperature increase is then added to the initial temperature to yield the temperature of the conductor at the end of the first-time step.

c) All heat terms are recalculated at this new temperature.

d) The subsequent increase in temperature for the next time step is calculated with the new heat terms. This temperature increase is added to the temperature at the end of the previous time step to yield the temperature of the conductor at the end of this time step.

e) This process is repeated until the conductor reaches steady state or until the desired overall time period is reached.

As the heat terms are assumed to be at the temperature at the beginning of the time step (rather than varying throughout it as it would in actuality), longer time steps will result in lower accuracy. Therefore, accuracy in the iterative transient calculation requires that the time step chosen be sufficiently small (< 1%) with respect to the thermal time constant. Therefore, time steps shall be 10 s or less. The calculation may be rerun with a smaller time step to check whether the calculated values change.

The rate of change in bare overhead conductor temperature is approximately exponential, with a thermal time constant that is on the order of 5 min to 20 min for typical transmission conductors where the longest time constant corresponds to the largest conductors. With reference to Figure 1, this implies that the conductor temperature increases to its final value in a time period of 15 min to 60 min. Transient ratings are therefore typically calculated for emergency currents persisting for 5 min and 30 min.

4.2.2 Transient thermal rating

The transient thermal rating is calculated by repeating the preceding calculations of $T_{\text{avg}}\,(t)$ over a range of I_f values, then selecting the I_f value that causes the conductor temperature to reach its maximum allowable value in the allotted time.

4.2.3 Fault current calculations

In the case of fault current calculations, the step increase in current is usually quite large (> 10 000 A), the corresponding time to reach maximum allowable temperature is typically short (< 1 s), and the maximum temperatures attained may approach the melting point of aluminum or copper. These calculations are assumed adiabatic because the heat loss by convection and radiation during such short times is assumed negligible in comparison to the heat stored in the conductor. Therefore, for fault current calculations the same procedure is followed as in 4.2.1 except the external heat loss and heat gain terms (q_c, q_r, and q_s) are assumed to be zero. With non-homogeneous conductors, such as ACSR, the heat generation in the lower conductivity steel core is much lower than in the surrounding aluminum strand layers and therefore for fault current calculations the heat capacity of the steel core is ignored.

4.3 Dynamic case (time-varying weather and current)

The temperature of an overhead power conductor is constantly changing in response to changes in electrical current and weather. The thermal model described in this standard may be applied to this case. To do this a transient analysis is performed for a time interval during which the line current and weather parameters (wind speed and direction, ambient temperature, etc.) are assumed to remain constant and equal to their average values during the time interval.

Conductor temperature is calculated using the same methodology as the transient case for individual time steps for the duration of the time interval (where the time step is much smaller than the time interval). The change in conductor temperature, ΔT_{avg}, during the time step, Δt, is calculated using the non-steady-state heat balance equation [Equation (2) in 4.4.2]. The conductor temperature at the end of the time step is simply the initial temperature plus the change in temperature. Through a series of such time steps, the conductor temperature is calculated at the end of each interval, thus producing an approximation to the conductor temperature as it varies over time with the line current and weather conditions.

As for the transient calculation, the accuracy of the resulting dynamic temperature calculation requires that the time step chosen be sufficiently small (10 s or less) with respect to the thermal time constant.

4.4 Formulas

4.4.1 Steady-state heat balance

$$q_c + q_r = q_s + I^2 \times R(T_{\text{avg}}) \tag{1}$$

$$I = \sqrt{\frac{q_c + q_r - q_s}{R(T_{\text{avg}})}} \tag{2}$$

4.4.2 Non-steady-state heat balance

$$q_c + q_r + mC_p \times \frac{dT_{\text{avg}}}{dt} = q_s + I^2 \times R(T_{\text{avg}}) \tag{3}$$

$$\frac{dT_{\text{avg}}}{dt} = \frac{1}{mC_p}\left[R(T_{\text{avg}}) \times I^2 + q_s - q_c - q_r\right] \tag{4}$$

$$\Delta T_{\text{avg}} = \frac{1}{mC_p}\left[R(T_{\text{avg}}) \times I^2 + q_s - q_c - q_r\right] \times \Delta t \tag{5}$$

4.4.3 Convective heat loss, q_c

Convective heat loss is customarily divided into two types: natural convection and forced convection. Natural convection, or free convection, occurs during still air conditions, where, in a continuous process, cool air surrounding the hot conductor is heated and rises, and is replaced by cool surrounding air. Forced convection occurs when blowing air moving past the conductor carries the heated air away. Natural convection has low cooling power compared to forced convection, being equivalent to forced convection at a wind speed of less than 0.2 m/s (0.6 ft/s).

Many researchers have measured the heat loss by convection and the results are well documented in the heat transfer literature. The forced and natural convection heat loss equations described in this subclause of the standard use the equations for convection from cylinders based upon extensive wind tunnel measurements as recommended by McAdams [B28]. McAdams recommends calculating both the natural convection heat loss and the forced convection heat loss and using the greater of the two values.

For forced convection, McAdams based his recommended curve on the reported work of many researchers. He plotted their results, which were in very close agreement, and developed coordinates for a recommended cooling curve. Equation (9) and Equation (10) are curve fits to his recommended coordinates. Because of the huge range of Reynolds numbers in his curve, he fit the curve in the two sections and recommended that the larger of the two values be used for forced convection heat loss. Therefore, as recommended by McAdams, convection shall be calculated by all three equations [natural convection, Equation (7) or Equation (8), and both forced convection Equation (9) and Equation (10)] and the largest value shall be used.[12] In the case of zero wind speed, only natural convection is present and force convection can be ignored.

$$q_c = \text{MAX}(q_{cn}, q_{c1}, q_{c2}) \tag{6}$$

[12]It has been argued that at low wind speeds, the convection cooling rate should be calculated by using a vector sum of the wind speed and a "natural" wind speed [B29]. However, to be conservative it is recommended that only the larger of the forced convection and natural convection heat loss rates should be used.

4.4.3.1 Natural convection, q_{cn}

With zero wind speed ("still air"), natural convection occurs, where the rate of heat loss is:

$$q_{cn} = 3.645 \times p_f^{0.5} \times D_0^{0.75} \times (T_s - T_a)^{1.25} \qquad \text{W/m} \tag{7}$$

$$q_{cn} = 1.825 \times p_f^{0.5} \times D_0^{0.75} \times (T_s - T_a)^{1.25} \qquad \text{W/ft} \tag{8}$$

4.4.3.2 Forced convection, q_{c1}, and q_{c2}

Equation (9) is a fit to the low Reynolds number end of the curve and Equation (10) is fit to the high Reynolds number end of the curve in McAdams [B28]. Equation (9) is correct at low winds but underestimates forced convection at high wind speeds. Equation (10) is correct at high wind speeds but underestimates forced convection at low wind speeds. Therefore, as shown in Equation (6), at any wind speed forced convective heat loss shall be calculated with both equations, and the largest of the three (either the two calculated forced convection heat loss rates or the natural convection heat loss rates) shall be used.

$$q_{c1} = K_{\text{angle}} \times [1.01 + 1.35 \times N_{Re}^{0.52}] \times k_f \times (T_s - T_a) \qquad \text{W/m or W/ft} \tag{9}$$

$$q_{c2} = K_{\text{angle}} \times 0.754 \times N_{Re}^{0.6} \times k_f \times (T_s - T_a) \qquad \text{W/m or W/ft} \tag{10}$$

The forced convection heat loss equations are valid over a large range of variables: [B35]

Table 2—Valid range of forced convection variables

Variable	SI units	US units
Diameter	0.01 mm-150 mm	3E^{-05} ft-0.49 ft
Air velocity	0 m/s-19 m/s	0 ft/s-62 ft/s
Air temperature	15 °C-260 °C (60 °F-500 °F)	
Wire temperature	21 °C-1004 °C (70 °F-1840 °F)	
Air pressure	40.5 kPa-405 kPa	0.4 atm-4.0 atm

Equation (9) and Equation (10) have not been validated outside of the range in Table 2. For air temperature and conductor temperature outside of the range in Table 2, additional caution is recommended to help ensure conservative line ratings.

4.4.3.3 Reynolds number, N_{Re}

$$N_{Re} = \frac{D_0 \times \rho_f \times v_w}{\mu_f} \tag{11}$$

Reynolds number is dimensionless and the use of consistent units (e.g., SI units or US units) is necessary to ensure that the resulting value is dimensionless.

4.4.3.4 Mean film temperature, T_{film}

Air density, ρ_f, dynamic viscosity of air, μ_f, and thermal conductivity of the air, k_f, are calculated at the mean film temperature of the conductor boundary layer:

$$T_{\text{film}} = \frac{T_s + T_a}{2} \tag{12}$$

4.4.3.5 Dynamic viscosity of air, μ_f

Dynamic viscosity of air, μ_f, is calculated by Equation (13) or Equation (14).

$$\mu_f = \frac{1.458 \times 10^{-6} \times (T_{\text{film}} + 273)^{1.5}}{T_{\text{film}} + 383.4} \qquad \text{kg/m–s or N–s/m}^2 \tag{13}$$

$$\mu_f = \frac{9.806 \times 10^{-7} \times (T_{\text{film}} + 273)^{1.5}}{T_{\text{film}} + 383.4} \qquad \text{lb/ft–s} \tag{14}$$

NOTE—$1 \times \text{N}-\text{s/m}^2 = 1\,\text{kg/m}-\text{s}$in SI units.

4.4.3.6 Air density, ρ_f

Air density, ρ_f, is calculated by Equation (15) or Equation (16). Air density is dependent upon conductor elevation above sea level, H_e, and mean film temperature T_{film}. The highest elevation applicable at the location of the line shall be used as this gives the most conservative results.

$$\rho_f = \frac{1.293 - 1.525 \times 10^{-4} \times H_e + 6.379 \times 10^{-9} \times H_e^{\,2}}{1 + 0.00367 \times T_{\text{film}}} \qquad \text{kg/m}^3 \tag{15}$$

$$\rho_f = \frac{0.080695 - 2.901 \times 10^{-6} \times H_e + 3.7 \times 10^{-11} \times H_e^{\,2}}{1 + 0.00367 \times T_{\text{film}}} \qquad \text{lb/ft}^3 \tag{16}$$

4.4.3.7 Thermal conductivity of air, k_f

Thermal conductivity of air, k_f, is calculated by Equation (17) or Equation (18).

$$k_f = 2.424 \times 10^{-2} + 7.477 \times 10^{-5} \times t_{\text{film}} - 4.407 \times 10^{-9} \times t_{\text{film}}^2 \qquad \text{W/m} \times {}^\circ\text{C} \tag{17}$$

$$k_f = 7.388 \times 10^{-3} + 2.279 \times 10^{-5} \times t_{\text{film}} - 1.343 \times 10^{-9} \times t_{\text{film}}^2 \qquad \text{W/}ft \times {}^\circ\text{C} \tag{18}$$

4.4.3.8 Wind direction factor, K_{angle}

The convective heat loss rate, calculated with Equation (9) and Equation (10), shall include the wind direction factor, K_{angle}, where ϕ is the angle between the wind direction and the conductor axis (ϕ is between 0° and 90° where 90° is perpendicular to the conductor axis and 0° is parallel):

$$K_{\text{angle}} = 1.194 - \cos(\phi) + 0.194 \times \cos(2\phi) + 0.368 \times \sin(2\phi) \tag{19}$$

Alternatively, the wind direction factor may be expressed as a function of the angle, β, between the wind direction and perpendicular to the conductor axis. This angle is the complement of ϕ, and the wind direction factor becomes:

$$K_{\text{angle}} = 1.194 - \sin(\beta) - 0.194 \times \cos(2\beta) + 0.368 \times \sin(2\beta) \tag{20}$$

This is the form of the wind direction factor as originally suggested in [B13]. For perpendicular wind (90° to the conductor axis), K_{angle} is 1.

4.4.4 Radiated heat loss, q_r

When a bare overhead conductor is heated above the temperature of its surroundings, energy is transmitted by radiation to the surroundings. The rate at which the energy is radiated is dependent primarily on the difference in temperature between the conductor and its surroundings, which are assumed to be at ambient temperature. The surface condition of the conductor, its emissivity, e, also affects the radiative heat transfer. Radiation is described by the Stefan-Boltzmann law, relating the radiative energy transmission to the difference between the conductor surface temperature, T_s, and the surrounding temperature, T_a, expressed in absolute (Kelvin) degrees to the fourth power. The constant in Equation (21) and Equation (22) include the Stefan-Boltzmann constant and conversion factors to produce a result in the desired units.

$$q_r = 17.8 \times D_0 \times \varepsilon \times \left[\left(\frac{T_s + 273}{100}\right)^4 - \left(\frac{T_a + 273}{100}\right)^4\right] \qquad \mathrm{W/m} \tag{21}$$

$$q_r = 1.656 \times D_0 \times \varepsilon \times \left[\left(\frac{T_s + 273}{100}\right)^4 - \left(\frac{T_a + 273}{100}\right)^4\right] \qquad \mathrm{W/ft} \tag{22}$$

4.4.5 Solar heat gain, q_s

The sun provides heat energy to the conductor. As shown in Equation (23), the amount of solar heat energy delivered to the conductor depends on the conductor's projected area (A'), the conductor's surface condition (i.e., its absorptivity, α), the total solar and sky radiated heat adjusted for elevation at the specific date, time, and location (Q_{se}), and the angle of incidence of the sun to the conductor (θ). The total solar and sky radiated heat is dependent on the position of the sun, the solar constant (the amount of solar irradiance per area at the top of Earth's atmosphere) as well as atmospheric attenuation. This equation considers only direct solar heating and does not include diffuse or reflected solar heating.

The sun's position in the sky can be calculated by determining the altitude of the sun (the angle of the sun above the horizon, H_c), and the azimuth of the sun (the angle of the sun along the horizon with North defined as 0°, Z_c), for the specific date, time, and location. In addition to the position of the sun, the angle of incidence of the sun to the conductor is dependent on the orientation of the conductor as given by the azimuth of the line (the angle of the line with respect to North, Z_l). North-South lines have an azimuth of line of 0° while East-West lines have an azimuth of line of 90°. There is no solar heat gain during nighttime (from sunset to sunrise) and therefore the result of Equation (23) shall be set to a minimum of 0 (W/m or W/ft).

$$q_s = \alpha \times Q_{se} \times \sin(\theta) \times A' \qquad \mathrm{W/m\ or\ W/ft} \tag{23}$$

where:

$$\theta = \arccos\left[\cos(H_c) \times \cos(Z_c - Z_1)\right] \tag{24}$$

4.4.5.1 Solar altitude, H_c

The solar altitude, H_c, in degrees (or radians), is given by Equation (25), where inverse trigonometric function arguments are in degrees (or radians). Solar altitude can vary from 0° (at sunrise and sunset) to a maximum of 90° for solar noon near the equator. During nighttime (from sunset to sunrise) solar altitude is 0° therefore the result of Equation (25) shall be limited to a minimum of 0°.

$$H_c = \arcsin[\cos(Lat) \times \cos(\delta) \times \cos(\omega) + \sin(Lat) \times \sin(\delta)] \tag{25}$$

4.4.5.2 Hour angle, ω

The hour angle, ω, in degrees is the angular measurement of time from solar noon and is calculated by the number of hours from noon times 15° as shown in Equation (26). For example, 11:00 a.m. is −15° and 2:00 p.m. is +30°. Hour angle therefore can vary from −180° to +180°. Hour angle is calculated from solar noon (when the sun is at its peak in the sky) and the actual local clock time will differ.

$$\omega = (\text{Time} - \text{Noon}) \times 15° \tag{26}$$

4.4.5.3 Solar declination, δ

The solar declination (the angular distance of the sun from the Earth's equator), δ, in degrees, is:

$$\delta = 23.45 \times \sin\left[\frac{284 + N}{365} \times 360\right] \tag{27}$$

where N is the day of the year (e.g., N is 21 for January 21st) and the argument of the sin is in degrees. The equation is valid for all latitudes whether positive (northern hemisphere) or negative (southern hemisphere). Solar declination ranges between −23.45° and +23.45°. A solar declination of +23.45° occurs at the summer solstice for the northern hemisphere.

4.4.5.4 Solar azimuth, Z_c

The solar azimuth, Z_c, (in degrees), is:

$$Z_c = C + \arctan(\chi) \tag{28}$$

where

$$\chi = \frac{\sin(\omega)}{\sin(\text{Lat}) \times \cos(\omega) - \cos(\text{Lat}) \times \tan(\delta)} \tag{29}$$

The solar azimuth is either between 0° and 180° during the morning or between 180° and 360° during the afternoon. Therefore, to calculate solar azimuth the solar azimuth variable, χ, needs to be adjusted by a solar azimuth constant, C, depending on if the hour angle [ω, as calculated by Equation (26)] is less than 0 (i.e., morning) or greater than or equal to 0 (i.e., afternoon) as shown in Table 3:

Table 3—Solar azimuth constant, *C*, as a function of hour angle, *ω*, and solar azimuth variable, c

Hour angle, ω, degrees	C if $\chi \geq 0$ degrees	C if $\chi < 0$ degrees
$-180 \leq \omega < 0$	0	180
$0 \leq \omega < 180$	180	360

4.4.5.5 Total solar and sky radiated heat intensity at sea level, Q_s

The solar heating at the surface of the Earth is dependent on the amount of atmospheric attenuation which depends on cloud cover, atmospheric pollution, and moisture content of the air. The heat flux density received by a surface at sea level shall be calculated with Equation (30) using the coefficients for clear air quality shown in Table 4. These coefficients are for clear air free from cloud cover or atmospheric pollution and are therefore conservative. With suitable engineering judgment the coefficients for industrial air quality shown in Table 5 may be used instead. The industrial atmosphere coefficients in Table 5 assume increased atmospheric attenuation and therefore will result in less solar heating (making them less conservative than the clear air coefficients). As there is no solar heat gain during nighttime, the result of Equation (30) shall be limited to a minimum of 0 (W/m² or W/ft²).

$$Q_s = A + B \times H_c + C \times H_c^2 + D \times H_c^3 + E \times H_c^4 + F \times H_c^5 + G \times H_c^6 \quad (30)$$

Table 4—Polynomial coefficients for solar heat intensity as a function of solar altitude corresponding to clear atmosphere

Clear atmosphere	SI	US
A	−42.2391	−3.9241
B	63.8044	5.9276
C	−1.9220	-1.7856×10^{-1}
D	3.46921×10^{-2}	3.223×10^{-3}
E	-3.61118×10^{-4}	-3.3549×10^{-5}
F	1.94318×10^{-6}	1.8053×10^{-7}
G	-4.07608×10^{-9}	-3.7868×10^{-10}

Table 5—Polynomial coefficients for solar heat intensity as a function of solar altitude corresponding to industrial atmosphere

Industrial atmosphere	SI	US
A	53.1821	4.9408
B	14.2110	1.3202
C	6.6138×10^{-1}	6.1444×10^{-2}
D	-3.1658×10^{-2}	-2.9411×10^{-3}
E	5.4654×10^{-4}	5.07752×10^{-5}
F	-4.3446×10^{-6}	-4.03627×10^{-7}
G	1.3236×10^{-8}	1.22967×10^{-9}

4.4.5.6 Total solar and sky radiated heat intensity corrected for elevation, Q_{se}

The solar heat intensity at the earth's surface shall be corrected for elevation by Equation (31). The highest elevation applicable at the location of the line shall be used as this gives the most conservative results. For elevations at sea level, $K_{\text{solar}} = 1$ and $Q_{se} = Q_s$.

$$Q_{se} = K_{\text{solar}} \times Q_s \quad (31)$$

where

$$K_{\text{solar}} = A + B \times H_e + C \times H_e^2 \quad (32)$$

Table 6—Coefficients for solar flux elevation correction in Equation (32)

	SI	US
A	1	1
B	1.148×10^{-4}	3.500×10^{-5}
C	-1.108×10^{-8}	-1.000×10^{-9}

4.4.6 Conductor electrical resistance, $R(T_{avg})$

The electrical resistance of bare overhead conductor varies with the conductor cross-section areas, power frequency, current, and temperature. This standard does not include the calculation of electrical resistance therefore resistance at a high and low temperature value ($R(T_{high})$ and $R(T_{low})$ shall be obtained from other sources such as the *Aluminum Electrical Conductor Handbook* [B1] and the *Overhead Conductor Manual* [B32] or from the conductor manufacturer. Resistance at the average temperature of the aluminum wires, $R(T_{avg})$, shall then be calculated by linear interpolation from the high and low temperature values by Equation (33).

$$R(T_{avg}) = \left[\frac{R(T_{high}) - R(T_{low})}{T_{high} - T_{low}}\right] \times (T_{avg} - T_{low}) + R(T_{low}) \tag{33}$$

For T_{avg} less than or equal to 100 °C, a low temperature value of 25 °C and a high temperature value of 75 °C shall be used. For T_{avg} greater than 100 °C, a low temperature value 25 °C and a high temperature value of 200 °C shall be used unless otherwise directed by the conductor manufacturer. [13]

For ac resistance, the obtained resistance values shall include skin effect (see 4.4.6.2) and magnetic core effects (see 4.4.6.3). These effects are not applicable for dc resistance. Therefore, it is imperative that the high and low temperature values are at the same frequency as the intended conductor's power frequency. That is, to obtain dc resistance, dc resistance values for $R(T_{low})$ and $R(T_{high})$ in Equation (33) shall be used and to obtain 60 Hz ac resistance, 60 Hz ac resistance values for $R(T_{low})$ and $R(T_{high})$ in Equation (33) shall be used.

At a frequency of 60 Hz, at temperatures of 25 °C to 75 °C, the [B1] and [B32] list tabulated values of 60 Hz electrical ac resistance for most sizes and types of bare overhead conductors. These references also provide dc resistance at 20 °C. Conductor manufacturers also typically provide such resistance values for their conductors. The conductor ac resistance at any conductor temperature shall include skin effect (see 4.4.6.2), and, for one- and three-layer steel-core conductors, magnetic core effects (see 4.4.6.3).

Direct calculation of resistance for a given temperature and the impact of radial thermal gradients within the conductor is not required by this standard; however, they may be included for increased accuracy. When radial thermal gradient is neglected, the average temperature of the aluminum wires, T_{avg}, is assumed equal to the surface temperature, T_s, and therefore the core temperature, T_{core}, as well. See 4.5 for further discussion.

4.4.6.1 Limitations of linear interpolation of ac resistance

Since the resistivity of most common metals used in overhead conductors increases somewhat faster than linearly with temperature, the resistance calculated by Equation (33) will be somewhat high (and thus conservative for rating calculations) so long as conductor temperature is between T_{low} and T_{high}. If the conductor temperature exceeds T_{high}, however, the calculated resistance will be somewhat low (and thus non-conservative for rating calculations).

Therefore, to limit the amount of extrapolation, in accordance with 4.4.6 for T_{avg} less than or equal to 100 °C, a low temperature value of 25 °C and a high temperature value of 75 °C shall be used. For T_{avg} greater than 100 °C, a low temperature value 25 °C and a high temperature value of 200 °C shall be used unless otherwise directed by the conductor manufacturer.

4.4.6.2 Skin effect component of ac resistance

The flow of ac current within a metal conductor tends to be concentrated toward the surface (or "skin") of the conductor due to internal magnetic flux increasing the inductance (and therefore reactance) at the center of the

[13]Given their approximate nature, this simple linear equation is often used to estimate resistance for conductor temperatures much higher than T_{high} for fault current calculations.

conductor thereby resulting in current being concentrated at the surface where reactance is lower. This “skin effect” is frequency-dependent and is magnified at higher frequencies. At 60 Hz, the increase in resistance due to skin effect in an aluminum conductor of overall diameter 30 mm is on the order of 1% to 2%. For larger conductors such as 1092 mm^2 (2156 kcmil) Bluebird ACSR, which has an outside diameter of 44.8 mm (1.762 in), the increase in resistance due to skin effect is on the order of 8%.

When calculating ac resistance, the values for $R(T_{\text{low}})$ and $R(T_{\text{high}})$ in Equation (33) shall include the increase in resistance from the skin effect. The resistance values in the referenced handbooks ([B1] and [B32]) include skin effect for all types of conductor. When calculating the resistance for a dc conductor application the skin effect does not occur and the corresponding dc resistance shall be substituted for the $R(T_{\text{low}})$ and $R(T_{\text{high}})$ in Equation (33).

4.4.6.3 Magnetic core effect on ac resistance

Within steel-core conductors such as ACSR and ACSS, the flow of ac current is primarily through the aluminum strands. Since the helically-wound aluminum strands surround the steel, magnetic flux is generated in the steel core, much like a solenoid, increasing with the magnitude of current. As a result, the resistance of the conductor will depend not only on the temperature of the conductor, but also on the magnitude of current. This adds additional uncertainty for electrical resistance when performing linear interpolation per Equation (33). To help reduce this effect, $R(T_{\text{low}})$ and $R(T_{\text{high}})$ in Equation (33) shall include the magnetic core effect for steel core conductors with an odd number of aluminum layers (e.g., single- and three-layer ACSR and ACSS). The single-layer ac resistance values provided in [B1] and [B32] include the magnetic core effect; however, as the effect varies with current these values are only accurate at the assumed weather conditions used for the calculation and some variance between sources is expected due to differences in assumptions.

The core’s impact on conductor resistance depends on the construction of the bare stranded conductor. For steel-core conductors with an even number of aluminum layers (e.g., Drake/ACSR, Ibis/ACSR, and Bluebird/ACSR) the magnetic core flux produced by the current in each layer essentially cancels, and level of magnetic flux in the core is quite low. No correction is necessary in this case.

For single-layer steel-core conductors, the magnetic flux in the steel core is quite high and the resulting magnetic hysteresis and eddy current losses in the core can increase the effective resistance by as much as 20% at high current levels.

For three-layer steel core conductors, there is partial magnetic field cancelation in the steel core so that the losses in the core are much smaller, but the solenoid effect couples the layer currents making the current densities unequal and increasing the overall ac resistance of the conductor by as much as 5% at high current levels.

4.4.7 Conductor heat capacity, mC_p

Conductor heat capacity is defined as the product of specific heat, C_p, and mass per unit length, m. If the conductor consists of more than one material (e.g., ACSR), then the conductor heat capacity is equal to the sum of the heat capacities of the core and the outer strands, each defined in this way.

For a non-homogeneous conductor such as ACSR, most of the heat (98% to 99%) is generated in the aluminum strands and transferred to the steel core with an internal time constant of less than 1 min. For steady-state and transient thermal rating calculations with durations of 5 min to 30 min, the temperature of the conductor components (aluminum and steel) are assumed equal, and the heat capacity of the conductor can be calculated as the sum of the component heat capacities as shown in Equation (34):

$$m\,C_p = \sum m_i \times C_{Pi} \tag{34}$$

Values for the specific heat of common metals used in overhead conductors are listed in Table 7 [B5]. For materials not listed in the table, consult the conductor manufacturer.

Table 7—Specific heat (C_p) of common overhead conductor materials

Material	C_p (J/kg × °C)	C_p (J/lb × °C)
Aluminum	955	433
Copper	423	192
Galvanized or Mischmetal steel	476	216
Aluminum-clad steel[14]	534	242

4.5 Known omissions (optional)

While this standard incorporates the significant sources of heating and cooling for overhead conductor, it also recognizes that there are other factors or variances within the equations that are not included in this standard. While not required by this standard, these known omissions may be included if desired in order to increase accuracy which can be useful in circumstances such as when performing dynamic line ratings. If any of these effects are included when performing calculations to this standard than it shall be explicitly notated.

4.5.1 Calculation of electrical resistance (without interpolation)

As stated in 4.4.6, this standard is explicitly not a standard for the calculation of electrical resistance. However, electrical resistance may be calculated directly without interpolation for a given temperature incorporating the skin effect (4.4.6.2) and the magnetic core effects (4.4.6.3) as necessary for the conductor. More information and methodology can be found in [B7].

4.5.2 Radial thermal gradient

As the conductor is not uniform in temperature there exists some thermal gradient from the core to the surface of the conductor. This gradient is minimal at low operating temperatures but can increase significantly at high operating temperatures. For high temperature conductors with operating temperatures above 100 °C the radial thermal gradient may have a significant effect.

Annex C provides the equations necessary to calculate the radial thermal gradient. When radial thermal gradient is neglected, the average temperature of the aluminum wires, T_{avg}, is assumed equal to the surface temperature, T_s, and therefore the core temperature, T_{core}, as well. When not neglected, radial thermal gradient can be incorporated into Equation (1) by calculating the conductor electrical resistance [per Equation (33)] at the average conducting wire temperature rather than the surface temperature. Iterative calculations are required to properly determine the average wire temperature as the core temperature is dependent on the joule losses which is dependent on average wire temperature.

In those calculations where the core of the conductor is significantly hotter than the surface, the sags of the line will be higher, the core material may experience greater deterioration, and the resistance of the conductor may be greater than anticipated if all these calculations are based upon the conductor surface temperature. Therefore, in those cases it may be necessary to reduce the maximum surface temperature if the higher core temperature results in conductor deterioration or inadequate sag clearance.

[14]The specific heat of aluminum-clad steel depends on the aluminum-to-steel ratio. This is a typical value for aluminum-clad steel wire with a conductivity of 20.3% I.A.C.S.

4.5.3 Diffuse and reflected solar heating

The exact calculation of conductor solar heat gain per unit length can be quite complex (e.g., [B13]). There are three sources of solar heating: direct solar heating; diffuse solar heating; and reflection from the ground under the line. This standard considers only direct solar heating, which is the largest of the three sources, but allows for all times of day, all days of year, and all latitudes.

Although not included in the direct solar heat gain calculations of this standard, [B9] allows the inclusion of both diffuse and reflected solar heating. This calculation requires the specification of reflectance (albedo) for the ground under the line. For most applications, the method in this standard is conservative though solar reflectance may have some impact on thermal rating calculations for low conductor design temperatures (i.e., 60 °C or lower).

4.5.4 Precipitation cooling

To be conservative, this standard does not include the additional cooling that can occur from precipitation. In the case of precipitation cooling, an additional heat sink term for the evaporation heat flux, q_e, can be added to the overall heat balance in Equation (1) to give:

$$q_c + q_r + q_e = q_s + I^2 \times R(T_{\text{avg}}) \tag{35}$$

This evaporation term accounts for heating the water to the minimum of the conductor temperature or the boiling temperature, and the latent heat of evaporation. The evaporation heat flux is given by [B33]:

$$q_e = m\left[L_e(T_e) + c_{p,w}(T_e - T_a)\right] \tag{36}$$

Where,

m is the mass of evaporating water,
L_e is the latent heat of evaporation,
T_e is the evaporation temperature,
T_a is the ambient temperature, and
$c_{p,w}$ is the specific heat of water

A correction to this term can be made to include the latent heat of fusion if the precipitation is snow. See [B33] for more detail on this calculation.

This equation makes two non-conservative assumptions that should be studied before using in applications. The first is that the heat transfer is efficient enough to evaporate water (or snow) at the same rate that the precipitation occurs. The second is that the downwards mass flux density of the precipitation impinge and collect on the surface of the conductor—i.e., no splashing, shedding, or dripping of the liquid occurs. The potential gain averages 163 A over periods of precipitation on a 35 mm conductor—about a 10% increase over the standard method.

4.6 Sample calculations

4.6.1 Steady-state thermal rating

The following example of calculating the steady-state thermal rating given a maximum allowable conductor temperature is intended to demonstrate the use of the equations discussed in this standard and, hopefully, yield some insight into the calculation process. It is important to note that the choice of weather conditions and conductor characteristics is for illustrative purposes only and does not constitute a recommendation of suitably conservative weather conditions for line thermal rating calculations. In addition, the number of significant digits does not indicate the accuracy of the equation.

4.6.1.1 Problem statement

Find the steady-state thermal rating (ampacity) for a 795 kcmil 26/7 Drake ACSR conductor, under the following conditions:

a) Wind speed $(V_w) = 0.61\ \text{m/s}\ (2.0\ \text{ft/s})$
b) Wind direction $(\phi) = 90°$ (perpendicular) to the conductor.
c) Emissivity $(\varepsilon) = 0.8$
d) Solar absorptivity $(\alpha) = 0.8$
e) Ambient air temperature $(T_a) = 40\ °\text{C}$
f) Maximum allowable conductor temperature = 100 °C
   Average temperature of aluminum strand layers $(T_{avg}) = 100\ °\text{C}$
   Conductor surface temperature$(T_s) = 100\ °\text{C}$
g) Conductor outside diameter$(D_0) = 28.14\ \text{mm}\ (1.108\ \text{in})$
h) Conductor ac resistance $\left[R(T_{\text{low}})\ \text{and}\ R(T_{\text{high}})\right] =$
   $R(25\ °\text{C}) = 7.283 \times 10^{-5}\ \Omega/\text{m}\ (2.220 \times 10^{-5}\ \Omega/\text{ft})$
   $R(75\ °\text{C}) = 8.688 \times 10^{-5}\ \Omega/\text{m}\ (2.648 \times 10^{-5}\ \Omega/\text{ft})$
i) The line runs in an east to west direction so azimuth of line, $Z_1 = 90°$
j) Latitude $(Lat) = 30°$ North
k) The atmosphere is clear
l) Time and day of the year = 11:00 a.m. on June 10 $(N = 161)$
m) Line elevation $(H_e) = 0\ \text{m}\ (0\ \text{ft})$

4.6.1.2 Air and conductor properties

$$\text{D}_0 = 28.14\ \text{mm} = 0.02814\ \text{m}$$
$$T_S = 100\ °\text{C}$$
$$T_a = 40\ °\text{C}$$
$$T_{\text{film}} = \frac{100 + 40}{2} = 70\ °\text{C}$$
$$\rho_f = 1.029\ \text{kg/m}^3$$
$$\mu_f = 2.043 \times 10^{-5}\ \text{kg/m} - s\ \text{or}\ \text{N} - \text{s/m}^2$$
$$k_f = 0.02945\ \text{W/m} - °\text{C}$$

$$D_0 = 1.108\ \text{in} = 0.09233\ \text{ft}$$
$$T_S = 100\ °\text{C}$$
$$T_a = 40\ °\text{C}$$
$$T_{\text{film}} = \frac{100 + 40}{2} = 70\ °\text{C}$$
$$\rho_f = 0.0642\ \text{lb/ft}^3$$
$$\mu_f = 1.374 \times 10^{-5}\ \text{lb/ft} - \text{s}$$
$$k_f = 0.008977\ \text{W/ft} - °\text{C}$$

4.6.1.3 Natural and forced convection heat loss, q_c

The natural convection heat loss is calculated by means of Equation (7) or Equation (8):

$$q_{cn} = 3.645 \times \rho_f^{0.5} \times D_0^{0.75} \times (T_s - T_a)^{1.25}$$
$$q_{cn} = 3.645 \times (1.029)^{0.5} \times (0.02814)^{0.75} \times (100 - 40)^{1.25}$$
$$q_{cn} = 42.42\ \mathrm{W/m}$$

$$q_{cn} = 1.825 \times \rho_f^{0.5} \times D_0^{0.75} \times (T_s - T_a)^{1.25}$$
$$q_{cn} = 1.825 \times (0.0642)^{0.5} \times (0.09233)^{0.75} \times (100 - 40)^{1.25}$$
$$q_{cn} = 12.93\ \mathrm{W/ft}$$

Since the wind speed is greater than zero, the forced convection heat loss is calculated according to both Equation (9) and Equation (10) and compared to the natural convection heat loss. The largest of the heat losses due to both natural and forced convection is used to calculate the thermal rating. As the wind is perpendicular to the conductor axis $K_{\mathrm{angle}} = 1$.

The Reynolds number is dimensionless and is calculated according to Equation (11). Reynolds number is dimensionless and therefore the same in either set of units (with the exception of slight differences due to rounding).

$$N_{\mathrm{Re}} = (D_0 \times \rho_f \times V_w)/\mu_f$$
$$N_{\mathrm{Re}} = \frac{0.02814 \times 1.029 \times 0.61}{2.043 \times 10^{-5}}$$
$$N_{\mathrm{Re}} = 865$$

$$N_{\mathrm{Re}} = \frac{0.09233 \times 0.0642 \times 2.0}{1.374 \times 10^{-5}}$$
$$N_{\mathrm{Re}} = 863$$

The forced convection heat loss is calculated in accordance with Equation (9) and Equation (10):

$$q_{c1} = K_{\mathrm{angle}} \times [1.05 + 1.35 \times N_{\mathrm{Re}}^{0.52}] \times k_f \times (T_s - T_a)$$
$$q_{c1} = 1 \times [1.01 + 1.35 \times 865^{0.52}] \times 0.02945 \times (100-40)$$
$$q_{c1} = 82.10\ \mathrm{W/m}$$

$$q_{c1} = 1 \times [1.01 + 1.35 \times 865^{0.52}] \times 0.008977 \times (100-40)$$
$$q_{c1} = 25.00\ \mathrm{W/ft}$$

$$q_{c2} = K_{\mathrm{angle}} \times 0.754 \times N_{\mathrm{Re}}^{0.6} \times k_f \times (T_s - T_a)$$
$$q_{c2} = 1 \times 0.754 \times 865^{0.6} \times 0.02945 \times (100-40)$$
$$q_{c2} = 77.06\ \mathrm{W/m}$$

$$q_{c2} = 1 \times 0.754 \times 863^{0.6} \times 0.008977 \times (100-40)$$
$$q_{c2} = 23.46\ \mathrm{W/ft}$$

As instructed in 4.4.3 and per Equation (6), the largest of the calculated three heat losses is selected. From the results shown, q_{c1} is greater than both q_{cn} and q_{c2} therefore $q_c = q_{c1}$.

$$q_c = 82.10\ \mathrm{W/m} \qquad q_c = 25.00\ \mathrm{W/ft}$$

4.6.1.4 Radiated heat loss, q_r

Radiated heat loss is calculated in accordance with Equation (21) or Equation (22) as follows:

$$q_r = 17.8 \times D_0 \times \varepsilon \times \left[\left(\frac{T_s + 273}{100}\right)^4 - \left(\frac{T_a + 273}{100}\right)^4\right]$$
$$q_r = 17.8 \times 0.02814 \times 0.8 \times \left[\left(\frac{100 + 273}{100}\right)^4 - \left(\frac{40 + 273}{100}\right)^4\right]$$
$$q_r = 39.11\ \mathrm{W/m}$$

$$q_r = 1.656 \times D_0 \times \varepsilon \times \left[\left(\frac{T_s + 273}{100}\right)^4 - \left(\frac{T_a + 273}{100}\right)^4\right]$$
$$q_r = 1.656 \times 0.09233 \times 0.8 \times \left[\left(\frac{100 + 273}{100}\right)^4 - \left(\frac{40 + 273}{100}\right)^4\right]$$
$$q_r = 11.94\ \mathrm{W/ft}$$

4.6.1.5 Solar heat gain, q_s

The conductor is at 30° North latitude and the line is oriented East to West. In order to calculate the solar altitude, the day of the year, N, is needed. June 10th is the 161st day of the year; therefore, N = 161.

The solar declination for June 10 is given by Equation (27):

$$\delta = 23.45 \times \sin\left[\frac{284 + 161}{365} \times 360\right]$$

$$\delta = 23.45 \times \sin\left[\frac{284 + 161}{365} \times 360\right]$$

$$\delta = 23.0°$$

The solar altitude, H_c, is found from the latitude (30°), the solar declination (23.0°) and the hour angle, ω, (−15°) with the use of Equation (25):

$$H_c = \arcsin[\cos(Lat) \times \cos(\delta) \times \cos(\omega) + \sin(Lat) \times \sin(\delta)]$$

$$H_c = \arcsin[\cos(30°) \times \cos(23.0°) \times \cos(-15°) + \sin(30°) \times \sin(23.0°)]$$

$$H_c = 74.9°$$

The solar azimuth variable, χ, is needed in order to calculate the solar azimuth, referring to Equation (28). The azimuth variable χ is calculated in accordance with Equation (29):

$$\chi = \frac{\sin(\omega)}{\sin(Lat) \times \cos(\omega) - \cos(Lat) \times \tan(\delta)}$$

$$\chi = \frac{\sin(-15°)}{\sin(30°) \times \cos(-15°) - \cos(30°) \times \tan(23.0°)}$$

$$\chi = -2.24$$

The solar azimuth constant is found in Table 3. Since the hour angle is negative and the solar azimuth variable is negative, the solar azimuth constant, C, is 180°. Having determined both the solar azimuth variable χ and the constant C, the solar azimuth angle is calculated in accordance with Equation (28):

$$Z_c = C + \arctan(\chi)$$

$$Z_c = 180 + \arctan(-2.24)$$

$$Z_c = 114°$$

The solar heat flux (heat intensity), Q_s, for clear air at sea level is given by Equation (30) using the coefficients in Table 4 as this is clear atmosphere:

$$Q_s = A + B \times H_c + C \times H_c^2 + D \times H_c^3 + E \times H_c^4 + F \times H_c^5 + G \times H_c^6$$

$$Q_s = -42.2391 + 63.8044 \times 74.9 - 1.9220 \times 74.9^2 + 3.46921 \times 10^{-2} \times 74.9^3 - 3.61118 \times 10^{-4} \times 74.9^4 + 1.94318 \times 10^{-6} \times 74.9^5 - 4.07608 \times 10^{-9} \times 74.9^6$$

$$Q_s = 1027\,\mathrm{W/m^2}$$

$$Q_s = -3.9241 + 5.9276 \times 74.9 - 1.7856 \times 10^{-1} \times 74.9^2 + 3.223 \times 10^{-3} \times 74.9^3 - 3.3549 \times 10^{-5} \times 74.9^4 + 1.8053 \times 10^{-7} \times 74.9^5 - 3.7868 \times 10^{-10} \times 74.9^6$$

$$Q_s = 95.4\,\mathrm{W/ft^2}$$

As the elevation is at sea level, no further correction for elevation is necessary.

The effective angle of incidence of the solar rays with the conductor is calculated in accordance with Equation (24):

$$\theta = \arccos[\cos(H_c) \times \cos(Z_c - Z_l)]$$

$$\theta = \arccos[\cos(74.9°) \times \cos(114° - 90°)]$$

$$\theta = 76.2°$$

Finally, the solar heat gain is calculated in accordance with Equation (23):

$$q_s = \alpha \times Q_{se} \times \sin(\theta) \times A'$$

$$q_s = 0.8 \times 1027 \times \sin(76.2°) \times 0.02814$$

$$q_s = 22.45\,\mathrm{W/m}$$

$$q_s = 0.8 \times 95.4 \times \sin(76.2°) \times 0.09233$$

$$q_s = 6.843\,\mathrm{W/ft}$$

4.6.1.6 Resistance at 100 °C

As T_{avg} is 100 °C, a T_{low} of 25 °C and a T_{high} of 75 °C is used. Resistance at 100 °C is calculated in accordance with Equation (33):

$$R(T_{avg}) = \left[\frac{R(T_{\text{high}}) - R(T_{\text{low}})}{T_{\text{high}} - T_{\text{low}}}\right] \times (T_{avg} - T_{\text{low}}) + R(T_{\text{low}})$$

$$R(T_{avg}) = \left[\frac{8.688 \times 10^{-5} - 7.283 \times 10^{-5}}{75 - 25}\right] \times (100 - 25) + 7.283 \times 10^{-5}$$

$$R(T_{avg}) = 9.391 \times 10^{-5}\,\Omega/\mathrm{m}$$

$$R(T_{avg}) = \left[\frac{2.648 \times 10^{-5} - 2.220 \times 10^{-5}}{75 - 25}\right] \times (100 - 25) + 2.220 \times 10^{-5}$$

$$R(T_{avg}) = 2.862 \times 10^{-5}\,\Omega/\mathrm{ft}$$

4.6.1.7 Steady-state thermal rating

Summary of results from preceding calculations:

$$q_c = 82.10\,\text{W/m} \qquad q_c = 25.00\,\text{W/ft}$$
$$q_r = 39.11\,\text{W/m} \qquad q_r = 11.94\,\text{W/ft}$$
$$q_s = 22.45\,\text{W/m} \qquad q_s = 6.843\,\text{W/ft}$$
$$R(100\,°\text{C}) = 9.391 \times 10^{-5}\,\Omega/\text{m} \qquad R(100\,°\text{C}) = 2.862 \times 10^{-5}\,\Omega/\text{ft}$$

Finally, steady-state thermal rating is calculated Equation (2):

$$I = \sqrt{\frac{q_c + q_r - q_s}{R(T_{\text{avg}})}}$$

$$I = \sqrt{\frac{82.10 + 39.11 - 22.45}{9.391 \times 10^{-5}}} \qquad I = \sqrt{\frac{25.00 + 11.94 - 6.84}{2.862 \times 10^{-5}}}$$

$$I = 1025\,\text{A} \qquad I = 1025\,\text{A}$$

4.6.2 Steady-state conductor temperature

As stated in 4.1.2, the calculation of steady-state conductor temperature for a given electrical current, weather conditions, and conductor characteristics can be calculated by the equations of 4.4 by making repeated calculations of the conductor current for a given conductor temperature. Using the same conductor characteristics and weather conditions in the example in 4.6.1 and assuming an electrical current of 1000 A, a temperature of 97.5 °C is calculated with the use of numerical iteration methods available in commercial spreadsheet software.

4.6.3 Transient conductor temperature

Equation (5) can be used to calculate the conductor temperature resulting from a change in conductor current. Assuming that the conductor begins in the steady-state condition described in the example problem of 4.6.1, this equation can be used to calculate the initial change in conductor temperature that occurs as a result of a sudden change in line current from 1025 A to 1200 A while the weather conditions remain constant:

a) Assume for Drake/ACSR, the nominal mass per unit length of the steel core is $0.5126\,\text{kg/m}$ ($0.3445\,\text{lb/ft}$) and nominal mass per unit length of the aluminum is $1.116\,\text{kg/m}$ ($0.7499\,\text{lb/ft}$).

b) Assume that the increase in resistance is linear past 100 °C, that is, $R(200\,°\text{C}) = 1.220 \times 10^{-4}\,\Omega/\text{m}$.

c) Assume a time step of 10 s as recommended by 4.2.

d) The initial current is 1025 A therefore the initial heat loss terms are the same as the example in 4.6.1.

e) Initially assume that the heat loss terms are unchanged and all of the heat goes into storage in the conductor, raising its temperature.

The total heat capacity of the conductor, mC_p, is calculated in accordance with Equation (33) and Table 7:

$$mC_p = \sum m_i \times C_{Pi}$$

$$mC_p = 0.5126 \times 476 + 1.116 \times 955 \qquad mC_p = 0.3445 \times 216 + 0.7499 \times 433$$

$$mC_p = 1310\,\text{J/(m} \times °\text{C)} \qquad mC_p = 399.1\,\text{J/(ft} \times °\text{C)}$$

Substituting the heat flow and resistance values from the example problem into Equation (5) where the conductor is at a surface temperature of 100 °C and the current is 1025 A:

For SI units:

$$\boldsymbol{\Delta} T_{\text{avg}} = \frac{1}{m\,C_p}\left[R\left(T_{\text{avg}}\right) \times I^2 + q_s - q_c - q_r\right] \times \boldsymbol{\Delta t}$$

$$\Delta T_{\text{avg}} = \frac{1}{1310}[9.391 \times 10^{-5} \times 1200^2 + 22.45 - 82.10 - 39.11] \times 10$$

$$\Delta T_{\text{avg}} = 0.28\,°\text{C}$$

For US units:

$$\boldsymbol{\Delta} T_{\text{avg}} = \frac{1}{m\,C_p}\left[R\left(T_{\text{avg}}\right) \times I^2 + q_s - q_c - q_r\right] \times \boldsymbol{\Delta t}$$

$$\Delta T_{\text{avg}} = \frac{1}{399.1}[2.862 \times 10^{-5} \times 1200^2 + 6.843 - 25.00 - 11.94] \times 10$$

$$\Delta T_{\text{avg}} = 0.28\,°\text{C}$$

At the end of this 10 s time period after the step change from 1025 A to 1200 A, the conductor temperature has increased from 100 °C to 100.28 °C. To continue the conductor temperature calculation, the conductor resistance, convection, and radiation heat loss terms are recalculated for a temperature of 100.28 °C. The solar heat input is unchanged. Therefore, the new heat loss values are:

$$q_c = 82.47\,\text{W/m} \qquad q_c = 25.12\,\text{W/ft}$$

$$q_r = 39.34\,\text{W/m} \qquad q_r = 12.01\,\text{W/ft}$$

$$q_s = 22.45\,\text{W/m} \qquad q_s = 6.843\,\text{W/ft}$$

$$R(100.28\,°\text{C}) = 9.398 \times 10^{-5}\,\Omega/\text{m} \qquad R(100.28\,°\text{C}) = 2.864 \times 10^{-5}\,\Omega/\text{ft}$$

During the next 10 s, the change in conductor temperature is slightly less as shown in the following equation with the revised resistance and heat loss values:

For SI units:

$$\boldsymbol{\Delta} T_{\text{avg}} = \frac{1}{m\,C_p}\left[R\left(T_{\text{avg}}\right) \times I^2 + q_s - q_c - q_r\right] \times \boldsymbol{\Delta t}$$

$$\Delta T_{\text{avg}} = \frac{1}{1310}[9.398 \times 10^{-5} \times 1200^2 + 22.45 - 82.47 - 39.34] \times 10$$

$$\Delta T_{\text{avg}} = 0.27\,°\text{C}$$

For US units:

$$\boldsymbol{\Delta} T_{\text{avg}} = \frac{1}{m\,C_p}\left[R\left(T_{\text{avg}}\right) \times I^2 + q_s - q_c - q_r\right] \times \boldsymbol{\Delta t}$$

$$\Delta T_{\text{avg}} = \frac{1}{399.1}[2.862 \times 10^{-5} \times 1200^2 + 6.843 - 12.12 - 12.01] \times 10$$

$$\Delta T_{\text{avg}} = 0.27\,°\text{C}$$

As long as the weather conditions and higher line current (i.e., 1200 A) remain the same, the conductor temperature can be "tracked" by this method. Eventually, the conductor will reach a new steady-state temperature of 119.6 °C with the change in temperature decreasing during each subsequent 10 s as the conductor approaches steady-state. This is illustrated in Figure 2 which shows a plot of conductor temperature versus time. Also included in Figure 2 is an example where the final line current is 1400 A instead 1200 A. As shown, the rate of increase in conductor temperature after the current step and the final steady-state conductor temperature both increase as the final current increases. The variation in conductor temperature with time is approximately exponential.

It is important to notice that the method works equally well in calculating the conductor temperature response to a sudden change in weather conditions (e.g., a drop in the wind speed). It also works if the weather conditions and line current are different in every calculation interval but, of course, the conductor temperature will not approach a new steady-state value unless both the weather conditions and line current stabilize.

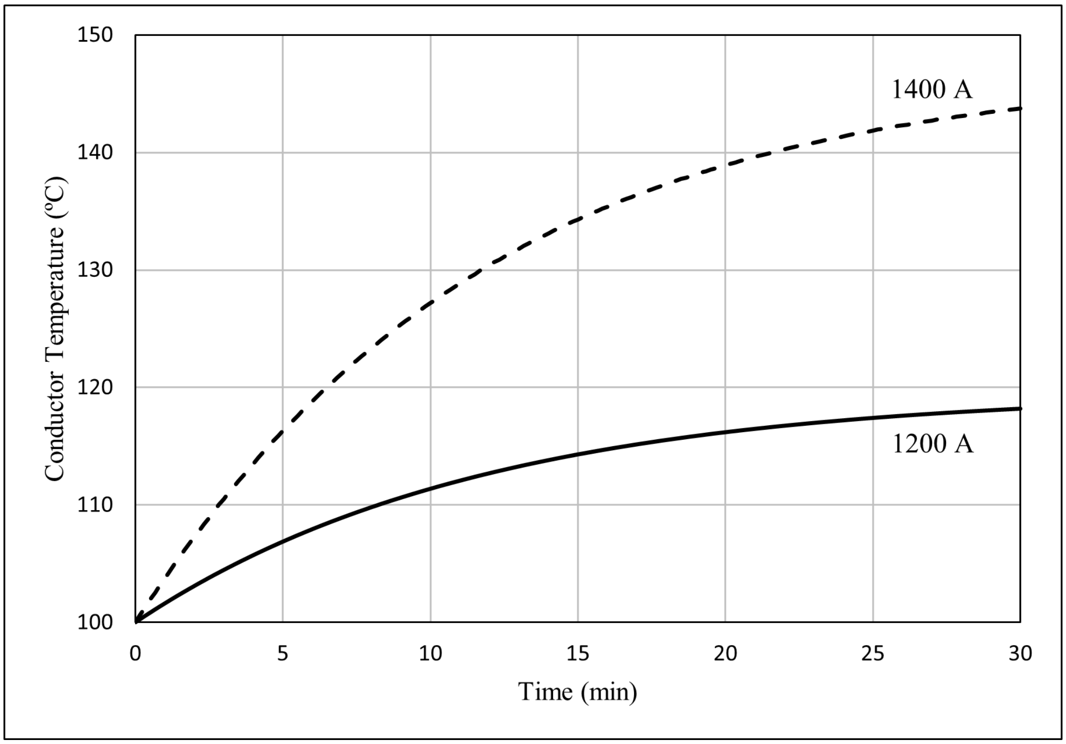

Figure 2—Transient temperature response to a step increase in current for Drake/ACSR under the assumptions of the example problem (10 s calculation interval)

4.6.4 Transient thermal rating

Often it is desirable to calculate the current that will yield a maximum allowable conductor temperature within a specific time period. To do so, the same procedure as 4.6.3 is followed; however, the current is now iterated as needed until the conductor temperature reaches the desired temperature after the prescribed time period. This is illustrated in the following sample problem.

Assume that the initial electrical current in a Drake/ACSR conductor is 1025 A steady-state at 100 °C at the same weather conditions and conductor parameters as described in the example problem of 4.6.1. It is then desired to calculate the conductor current that will yield a final temperature of 125 °C in 15 min (i.e., the 15 min transient thermal rating).

Referring to Figure 2, it can be seen that a final current of 1200 A causes the conductor temperature to increase from 100 °C to approximately 115 °C in about 15 min. From the same figure, one can also see that a final current of 1400 A causes the conductor temperature to reach 135 °C in about 15 min. Therefore, the 15 min transient thermal rating is between 1200 A and 1400 A for the given weather conditions.

Making use of commercial spreadsheet software and repeating the calculations of 4.6.3 for various current levels, it can be found that a current of 1312 A reaches 125 °C in 15 min. Note that this temperature would continue to rise after this 15 min period unless the current is reduced or the weather conditions become more favorable.

4.6.5 Dynamic conductor temperature (time-varying conductor current and weather conditions)

In the most general case thermal calculation, the line current and weather conditions vary over time as in the following example recorded for a transmission line with Drake/ACSR conductor.

For this example, consider 10 min time intervals. In this particular case, the weather conditions and the line current are averaged over each 10 min interval. For the first 10 min interval between midnight (00:00) and 10 min after midnight (00:10), the line current and weather conditions are assumed constant (while all other conditions and conductor properties are assumed to be the same as in the preceding example):

- Current (I) = 678 A
- Ambient air temperature (T_a) = 25 °C
- Solar heat gain (q_s) = $0.00\ \mathrm{W/m^2}(0.00\ \mathrm{W/ft^2})$

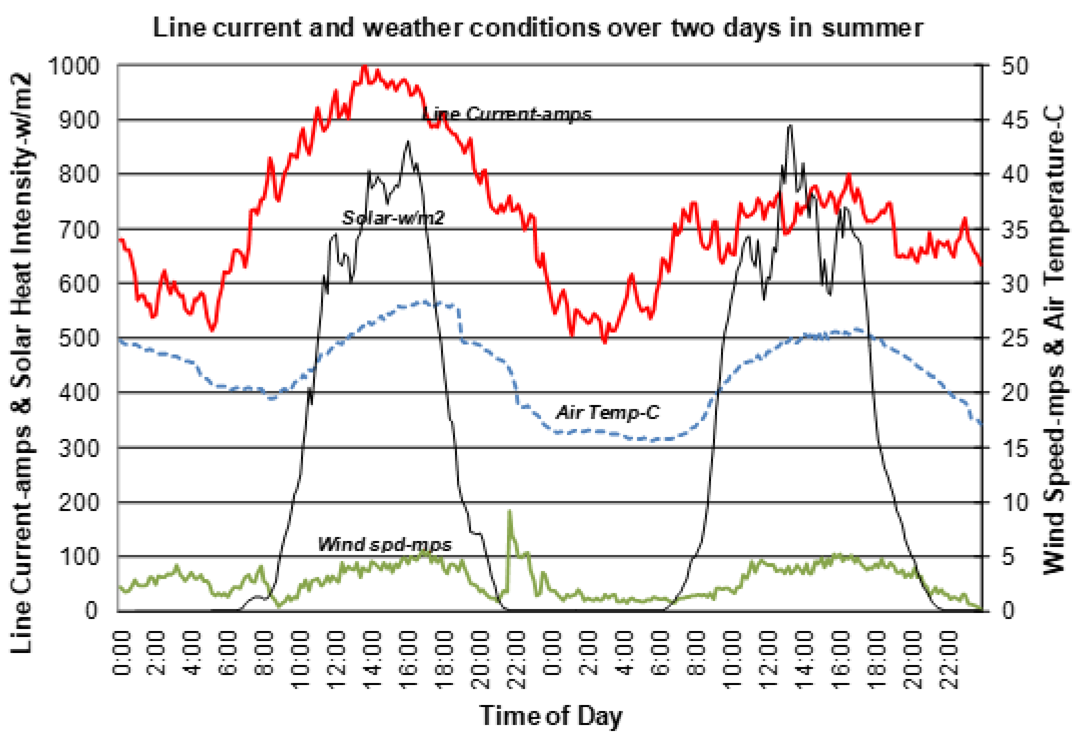

Figure 3—Example line current and weather conditions over a two-day period

– Wind velocity $(V_w) = 2.1\,\mathrm{m/s}(6.9\,\mathrm{ft/s})$

– Wind direction $(\phi) = 62°$ from conductor axis

To begin the process of tracking the conductor temperature over the two days for which the field data is available, assume that, at the beginning of the 10 min interval, the Drake/ACSR conductor is at 50 °C. Of course, it may be hotter or cooler than this depending on what the line current and weather conditions were previous to the field data test period, but the conductor temperature will converge to its true value after a number of 10 min intervals. For each 10 min interval, the change in conductor temperature will be based on the conductor temperature at the beginning of the interval and the line current and weather conditions during it.

Therefore, for this first interval, the terms in the heat balance will be calculated for the conductor temperature at 00:00 (i.e., 50 °C), a line current of 678 A, and the weather conditions listed in the preceding for the first 10 min time interval.

The thermal time constant of Drake/ACSR is approximately 10 min, depending on the wind speed and direction. For illustrative purposes, a time step of 1 min (60 s) will be used for this example. For the first 1 min time step, the change in temperature is determined by Equation (4) with the following heat balance terms:

$q_c = 63.31\,\mathrm{W/m}$ $\quad$ $q_c = 19.29\,\mathrm{W/ft}$

$q_r = 12.01\,\mathrm{W/m}$ $\quad$ $q_r = 3.670\,\mathrm{W/ft}$

$q_s = 0.00\,\mathrm{W/m}$ $\quad$ $q_s = 0.00\,\mathrm{W/ft}$

$R(50\,°\mathrm{C}) = 7.986 \times 10^{-5}\,\Omega/\mathrm{m}$ $\quad$ $R(50\,°\mathrm{C}) = 2.434 \times 10^{-5}\,\Omega/\mathrm{ft}$

Therefore, the change in temperature is calculated per Equation (5):

For SI units:

$$\Delta T_{\mathrm{avg}} = \frac{1}{m\,C_p}\left[R\left(T_{\mathrm{avg}}\right) \times I^2 + q_s - q_c - q_r\right] \times \Delta t$$

$$\Delta T_{\mathrm{avg}} = \frac{1}{1310}[7.986 \times 10^{-5} \times 678^2 + 0 - 63.31 - 12.01] \times 60$$

$$\Delta T_{\mathrm{avg}} = -1.77\,°\mathrm{C}$$

For US units:

$$\Delta T_{\mathrm{avg}} = \frac{1}{m\,C_p}\left[R\left(T_{\mathrm{avg}}\right) \times I^2 + q_s - q_c - q_r\right] \times \Delta t$$

$$\Delta T_{\mathrm{avg}} = \frac{1}{1310}[2.434 \times 10^{-5} \times 678^2 + 0 - 19.29 - 3.670] \times 60$$

$$\Delta T_{\mathrm{avg}} = -1.77\,°\mathrm{C}$$

The average conductor temperature therefore drops from 50 °C to 48.23 °C. During the next 1 min time step, the weather conditions and line current remain the same, but the conductor temperature used to calculate the heat balance terms drops to 48.23 °C.

After 10 such calculation intervals, the weather conditions and line current are adjusted to their next 10 min average values and the process repeated. Going through all of the available 10 min intervals indicated in Figure 3, the conductor temperature can be plotted as a function of time as shown in the following Figure 4.

In summary, the numerical heat balance method described in this standard works equally well when performing steady-state, transient, and dynamic calculations.

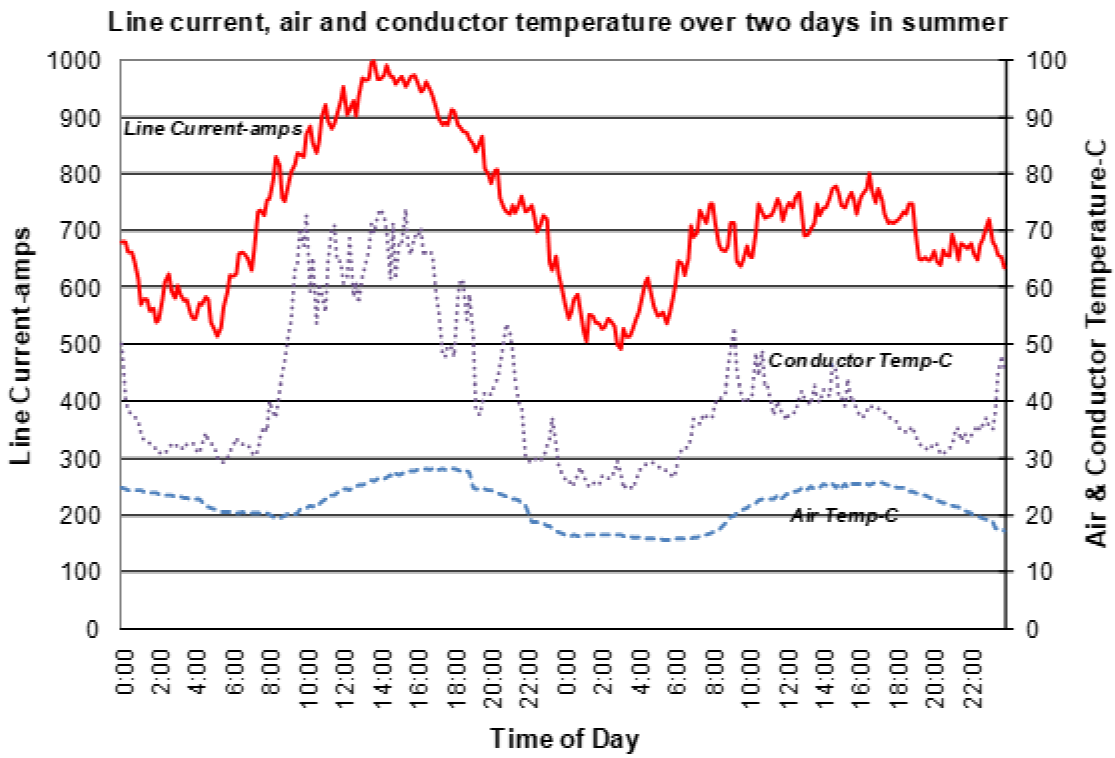

Figure 4—Conductor temperature calculated by the 738 numerical method, given time-varying line current and weather conditions

5. Input data

5.1 General

This standard explicitly does not recommend suitable weather conditions or conductor parameters for use in line rating calculations. This standard primarily concerns mathematical methods by which the temperature of a bare overhead conductor can be accurately calculated given the electrical current through it and the weather conditions immediately around it. However, it is recognized that even the most sophisticated calculation method can give inaccurate answers if the conductor parameters (resistance, emissivity and absorptivity), the solar heat intensity and solar angles, and the weather parameters (wind speed, wind direction, and air temperature) are inaccurate.

The selection of these conductor and weather parameters are discussed in this section though the actual selection of suitably conservative inputs is left to the user's engineering judgment.

5.2 Selection of weather conditions

The process of selecting conservative weather conditions are described in CIGRE [B8]. This reference provides guidance for four separate types of rating calculations:

a) In the absence of data from field rating studies, concurrent worst-case weather conditions should be used. The concurrent worst-case weather condition is one where all variables are considered in tandem rather than taken as the worst-case for each variable independently. For instance, the worst-case wind speed may not occur at the same time as the worst-case ambient temperature. Instead, the wind speed and ambient condition that, combined together, produce the worst-case scenario should be used instead.

b) If field rating studies are undertaken, the transmission line owner/operator may base the rating weather assumptions of selected lines or transmission regions upon such studies, provided that they are conducted in the actual transmission line environment, using the instrumentation methods recommended in [B8].

c) Ratings can be adjusted based on measured or forecasted ambient temperatures in the vicinity of the line. These are termed continually ambient-adjusted ratings and [B8] suggests that the choice of worst-case wind speeds should depend on the assumed air temperature.

d) The transmission line owner/operator may elect to use real time monitoring equipment for determining the line rating, provided that the monitoring equipment meets certain sensitivity, accuracy, and calibration requirements specified in [B8].

A more detailed discussion of the selection of weather conditions as provided in [B8] is included in Annex E.

5.3 Air density, viscosity, and conductivity

The density, viscosity, and thermal conductivity of air are used in the calculation of convection losses and can be obtained from the equations in 4.4.3. These air parameters are a function of the air temperature and the conductor surface temperature. Air density is also a function of elevation above sea level. It is recommended that the highest elevation that is applicable at the location of the line be selected, because this will give the most conservative results.

5.4 Conductor emissivity and absorptivity

Values for emissivity and absorptivity for some conductors can be obtained from the conductor manufacturer. Emissivity and absorptivity are generally correlated, with absorptivity assumed to be slightly higher than emissivity. Recent laboratory measurements of conductor samples by EPRI [B10][B19], support the use of an initial value of between 0.2 and 0.4 for bare aluminum that will gradually increase in most environments. The exact rate of increase depends on the density of atmospheric particulates and the line's operating voltage. Existing testing does not support use of values above 0.6 for aged conductors. These guidelines do not apply to conductors that are coated with special materials to increase emissivity, decrease absorptivity, or both.

Historically, in North America, values for thermal rating calculations have been either that both parameters are 0.5 or that both are in the range of 0.7 to 0.9. An incorrect emissivity or absorptivity value increases risks related to inaccurate ratings. Incorrect assumed values lead to large rating calculation errors at high conductor temperatures, but even at modest conductor temperatures (less than 100 °C), the total temperature error is small in magnitude but can represent an important portion of available capacity and influence the overall risk of exceeding design limits for a line.

At high conductor temperatures (greater than 150 °C), the value of emissivity has a larger impact on thermal rating because of increased radiation heat loss. At low conductor temperatures (less than 75 °C), the value of absorptivity has a larger impact on rating because of the importance of solar temperature rise.

When interpreting real-time monitor measurements made at normal line loadings, an incorrect absorptivity value can lead to large dynamic line rating calculation errors.

Where there is doubt as to what value of emissivity is to be assumed, measurements can be performed on conductor samples taken from the field.

5.5 Solar heat gain

Utility steady-state thermal line ratings are often calculated for the entire year or by season. For real-time rating calculations, the direct-beamed and sky or diffuse radiation may be input directly from data obtained from weather stations ([B12] and [B14]).

Solar heat input to a bare overhead conductor can cause a conductor temperature rise above air temperature of up to 20 °C in still air, though values are typically lower and on the order of 10 °C. The amount of heat gain will depend on conductor size, conductor surface condition, time of the year, and geographic location. At this time there is insufficient industry data to provide specific recommendations on conductor absorptivity without specific testing. In the absence of test data, the absorptivity can be taken as equal or slightly higher than the selected emissivity by adding a value of no more than 0.2.

5.6 Maximum allowable conductor temperature

It is important to distinguish between the maximum allowable conductor temperature used for line rating purposes and the actual conductor metal temperature. The maximum allowable conductor temperature is used to compute the maximum allowable line current based on the conductor properties and assumptions including the wind speed, ambient temperature, solar heat flux, and the conductor surface emissivity and absorptivity. However, the actual conductor temperature depends on how well the field conditions match these assumed environmental and conductor properties.

Overhead line conductor temperature needs to be managed to help prevent sag violations and to help avoid permanent damage to the conductor and/or accessories. Rating methods may under-estimate the metal temperature due to several factors including:

a) Current is assumed to be steady-state, but in reality current is generally variable. The conductor temperature reflects approximately the 15-min average of the current.

b) Wind is assumed to be steady state, but winds are typically gusty/variable, especially on sunny days when ground thermals are active. The conductor temperature reflects approximately the 15-min average of the wind speed.

The highest metal temperature occurs when the line current is near its rating, ambient temperature is at or near the value used to compute the rating, and convective heat loss is at or below the values used to compute the rating (due to lower wind speeds or a change in wind direction). The metal temperature at rated current can significantly exceed the maximum allowable temperature when the convective heat loss is lower than the assumed convective heat loss, and the ambient temperature is near the value used to compute the rating. The likelihood of these events simultaneously occurring is dependent on the assumptions used for line rating purposes and while these coincidences may be rare, they have occurred.

For conductors with full hard aluminum such as ACSR, elevated temperatures can cause annealing of the aluminum leading to significant loss-of-strength in the conductor. Annealing is a function of both exposure time and exposure temperature. The actual rate of annealing depends on the quantity and type of micro-impurities in the metal. The annealing rate is also affected by the processing history. Table 8 provides rule-of-thumb/order-of-magnitude annealing rate based on laboratory data for 1350-H19 aluminum strands. The actual conductor loss-of-strength will depend on the ratio of aluminum-to-steel. Note that annealing data comes from laboratory ovens where the temperature is calibrated and carefully controlled so that the metal temperature is stable. More information on elevated temperature operation of bare conductor can be found in IEEE Std 1283™ [B27].

Table 8—Rule-of-thumb annealing rate for 1350-H19 aluminum

Temperature	Time (hrs) for 5% loss of strength
75 °C	10 000
100 °C	1000
125 °C	100
150 °C	10
175 °C	1

It is common to use a higher maximum allowable conductor temperature for transient thermal rating calculations than for steady-state thermal rating calculations due to their time limitations. For fault calculations, the maximum allowable temperature is normally close to the melting point of the conductor material as durations are extremely short.

Annex A

(informative)

Bibliography

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Annex B

(informative)

Temperature calculation methods

In preparing this standard, the working group and previous working groups studied various methods used in calculating heat transfer and heat balance of bare overhead transmission line conductors included the following:

- House and Tuttle [B26]
- House and Tuttle, as modified by East Central Area Reliability (ECAR) [B38]
- Mussen, G. A. [B31]
- Pennsylvania-New Jersey-Maryland Interconnection (Davidson et al. [B11] and PJM Task Force [B16])
- Schurig and Frick [B34]
- Davis [B13]
- Morgan [B29]
- Black, Bush, Rehberg, and Byrd [B3], [B4], [B5]
- Foss, Lin, and Fernandez [B22]
- CIGRE Technical Brochure 207 [B9]

To differentiate between laminar and turbulent air flow, the House and Tuttle method [B22] uses two different equations for forced convection; the transition from one to the other is made at a Reynolds number of 1000. Because turbulence begins at some wind velocity and reaches its peak at some higher velocity, the transition from one curve to another is a curved line, not a discontinuity. The single transition value was selected as a convenience in calculating conductor ampacities.

The single transition value results in a discontinuity in current magnitude when this value is reached. Therefore, to avoid this discontinuity that occurs using the House and Tuttle method [B26], ECAR [B38] elected to make the change from laminar to turbulent air flow at the point where the curves developed from the two equations [Equation (9) and Equation (10)] cross. The equations for forced convection heat loss have an upper limit of application validity of a Reynolds number of 50 000 [B28] which is an order of magnitude higher than overhead transmission line conductors experience. For additional information on convection heat loss, see 4.4.3 of this standard.

The calculation method specified avoids certain simplifications that might be advisable where the speed or complexity of calculations is important.

In Davis [B13], the heat balance equation is expressed as a bi-quadratic equation that can be solved to give the conductor temperature directly. In Black and Rehberg [B5], and Wong et al. [B40], the radiation term is linearized and the resulting approximate linearized heat balance equation is solved using standard methods of linear differential equations. In Foss [B22], a somewhat more precise linearized radiation term is used to reduce the number of iterations required. These methods are computationally faster than the iterative method described in this standard; however, the algebraic expressions are more complex.

Annex C

(informative)

Radial thermal gradient [B18], [B34]

Since this standard was first published in 1986, the maximum conductor temperatures used to determine line ratings with conventional bare conductors has gradually increased from the range of 50 °C to 75 °C, to the range of 95 °C to 150 °C. In addition, high-temperature, low-sag conductors rated at from 150 °C to 250 °C have come into widespread use. As a result, the internal temperature difference between core and surface of conductors can no longer be neglected in all cases.

Bare transmission conductors are typically 12 to 50 mm in diameter, having one to four layers of helically stranded aluminum wires. High temperature conductors are designed to handle current densities as high as 5 A/mm^2 (2.5 A/kcmil) attaining surface temperatures as high as 250 °C. Heat generated in the inner layers of aluminum strands need be conducted to the surface layer in order to be dissipated and, since heat only flows from high to low temperature, it is reasonable to suspect that the core of the conductor may be somewhat hotter than the surface. Weather parameters have a direct effect on the surface temperature (as described in this standard) but only an indirect effect on the radial temperature gradient within the conductor. A reduction in the tension in the aluminum layers may be expected to reduce the contact pressure between the layers and increase any radial temperature gradient.

The magnitude of the radial temperature difference depends on a number of conductor parameters including:

— Conductor strand shape (round or trapezoidal or “Z” shaped).
— Magnitude of electrical current in aluminum layers.
— Electrical resistance of the aluminum strand layers.
— The number of layers of aluminum wires.
— The condition of aged conductor (i.e., corrosion and bird-caging).
— The contact area and pressure between aluminum layers.

Regardless of the conductor construction, the radial temperature difference is typically less than 5 °C when the current density is less than 1 A/mm^2 (2 A/kcmil) and the radial temperature difference may be neglected. This typically corresponds to ratings with a maximum conductor surface temperature of less than 100 °C.

At higher current densities, especially for large conductors with three or four layers of aluminum strands, radial temperature differences as large as 10 °C to 25 °C have been measured in laboratory tests.

Significant radial temperature differences are possible in any multi-layer aluminum conductor, whether having a steel core or not. Heat generation in the steel core of ACSR is normally less than 2% of the total Joule heat generated.

Because of the greater contact area between aluminum layers, the use of trapezoidal aluminum wires may reduce the radial temperature difference as long as the wires are under tension. However, once the aluminum wires have become slack, whether plastic elongation due to high ice loads or thermal elongation due to high temperature, the lack of contact pressure between aluminum layers is likely to lead to a significant radial temperature difference between the conductor core and surface.

CIGRE Technical Brochure 207, “Thermal behaviour of overhead conductors” [B9], includes an equation that allows the calculation of temperature difference if the “effective radial thermal conductivity” of the conductor is specified. For non-homogenous conductors with a core such as ACSR, the equation is:

$$T_{\text{core}} - T_s = \frac{I^2 \times R(T_{\text{avg}})}{2\pi \times k_{th}} \times \left[\frac{1}{2} - \frac{D_{\text{core}}^2}{D_0^2 - D_{\text{core}}^2} \times \ln\left(\frac{D_0}{D_{\text{core}}}\right)\right] \tag{C.1}$$

where

T_{core} is the temperature at the core-outer material interface

For all aluminum conductors, the equation for radial temperature difference is simpler as shown in Equation (C.2):

$$T_{\text{core}} - T_s = \frac{I^2 \times R(T_{\text{avg}})}{4\pi \times k_{th}} \tag{C.2}$$

where

T_{core} is the temperature at the center of the conductor

In Equation (C.1) and Equation (C.2), the resistance per unit length and the effective radial thermal conductivity shall be in consistent units. In SI units, resistance is in Ω/m and thermal conductivity is in $\mathrm{W/m-°C}$. In mixed US units, the resistance is in Ω/ft, and the effective radial thermal conductivity is in $\mathrm{W/ft-°C}$.

In this case, T_{avg} cannot be assumed to be equal to T_s and T_{core}. Therefore, T_{avg} is calculated as the average of these two values:

$$T_{\text{avg}} = \frac{T_{\text{core}} + T_s}{2} \tag{C.3}$$

When using Equation (C.1) and Equation (C.2), the resistance of the conductor is dependent upon T_{avg}, which is itself dependent on the temperature of the core as shown in Equation (C.3). Therefore, iteration is necessary to calculate the gradient. First, T_{avg} is assumed equal to T_s (and T_{core}). T_{core} is then calculated from Equation (C.1) or Equation (C.2). Then the new T_{avg} is calculated from Equation (C.3) and T_{core} is recalculated. This iterative process continues until T_{avg} no longer changes within a user defined tolerance. This process is achieved easily through the use of numerical methods in commercial spreadsheet software.

There is extensive technical literature that supports the existence of such radial temperature gradients at high current densities, but there is considerable variation in the recommended value for effective radial thermal conductivity, k_{th}. There remains no universal consensus and further research is ongoing to quantify the effective thermal conductivity considering the factors listed above. CIGRE [B6] states that for ACSR conductors below their “knee-point temperature” with tension in the aluminum strand layers, an effective thermal conductivity (k_{th}) of $2\ \mathrm{W/m-°C}$ appears to be reasonable. For ACSR conductors above their knee-point temperature and for other conductors where there is little or no tension in the aluminum strand layers, the use of an effective thermal conductivity equal to $1\ \mathrm{W/m-°C}$ is a more conservative value. EPRI test data suggest thermal conductivity is bound between 0.75 and $8.0\ \mathrm{W/m-°C}$ for conductors with round strands [B19].

C.1 Example including radial thermal gradients

Continuing the example of 4.5.1 and having determined the current in the conductor and the resistance, the radial temperature gradient can be calculated using Equation (C.1). For this example, assume that the effective radial thermal conductivity, k_{th}, is $1\ \mathrm{W/m-°C}$ $(0.305\ \mathrm{W/ft-°C})$ and the diameter of the core, D_{core}, for Drake/ACSR is 10.4 mm (0.408 in). The calculation of electrical resistance at T_{avg} requires an iterative calculation. For the first iteration, T_{avg} is assumed equal to the surface temperature, T_s.

$$T_{\text{core}} - T_s = \frac{I^2 \times R(T_{\text{avg}})}{2\pi \times k_{th}} \times \left[\frac{1}{2} - \frac{2^2_{\text{core}}}{D_0^2 - D^2_{\text{core}}} \times \ln\left(\frac{D_0}{D_{\text{core}}}\right)\right]$$

$$T_{\text{core}} - T_s = \frac{1025^2 \times 9.391 \times 10^{-5}}{2\pi \times I} \times \left[\frac{1}{2} - \frac{10.4^2}{24.14^2 - 10.4^2} \times \ln\left(\frac{28.14}{10.4}\right)\right]$$

$$T_{\text{core}} - T_s = 5.4\,°\text{C}$$

Therefore, with a T_s of 100 °C, T_{core} is equal to 105.4 °C. T_{avg} is then recalculated:

$$T_{\text{avg}} = \frac{T_{\text{core}} + T_s}{2}$$

$$T_{\text{avg}} = \frac{105.4 + 100}{2}$$

$$T_{\text{avg}} = 102.7\,°\text{C}$$

Thermal gradient is then recalculated with this new T_{avg}. Through the use of commercial spreadsheet software, it can be found that the temperature for this example quickly converges and T_{core} remains at a value of 105.4 °C at the end of the iterative process.

Annex D

(informative)

Conductor thermal time constant

The non-steady-heat balance Equation (3) cannot be solved analytically for conductor temperature as a function of time since certain of its terms are non-linear. Considering the equation term by term, it may be seen that the Joule heating term and the forced convection term are linear in regard to conductor temperature. The solar heating term is also linear since it is independent of conductor temperature. The radiation heat loss term and the natural convection (zero wind speed) term are both non-linear in regard to conductor temperature.

Several references [B5], [B22], and [B29] describe methods of approximating the radiation cooling [Equation (21) or Equation (22)] as a linear function of temperature. Doing so yields a linear non-steady-state heat balance equation of the form:

$$\frac{d}{dt}(T_c - T_a) = K_1 \times (T_c - T_a) = K_2 \times I^2 \tag{D.1}$$

For a step change in electrical current, the solution of the linearized non-steady-state heat balance equation is:

$$T_c(t) = T_i + \left(T_f - T_i\right) \times \left(1 - e^{-t/\tau}\right) \tag{D.2}$$

where

τ is the thermal time constant calculated per Equation (D.3)

$$\tau = \frac{\left(T_f - T_i\right) \times m\,C_p}{R(T_c) \times \left(I_f^2 - I_i^2\right)} \tag{D.3}$$

where

T_i is the steady state conductor temperature prior to the step increase in current and
T_f is the steady-state conductor temperature which occurs long after the step increase in current

Conductor resistance, $R(T_c)$ is that corresponding to the average conductor temperature, $\left(T_i + T_f\right)/2$.

Consider the exponential change in conductor temperature shown in Figure D.1. This is the "1200 amp" curve shown previously in Figure 2. The current undergoes a step change from 1025 A to 1200 A. The initial conductor temperature is 100 °C. The final conductor temperature is 119.5 °C. If the average conductor temperature is taken as 109.8 °C, the resistance of the Drake/ACSR conductor is $9.666 \times 10^{-5}\,\Omega/m(2.946 \times 10^{-5}\,\Omega/\text{ft})$. The time constant is:

$$\tau = \frac{\left(T_f - T_i\right) \times m\,C_p}{R(T_c) \times \left(I_f^2 - I_i^2\right)}$$

$$\tau = \frac{(119.5 - 100) \times 1310}{9.666 \times 10^{-5} \times (1200^2 - 1025^2)}$$

$$\tau = 679\ \text{s} = 11.3\ \text{min}$$

Alternatively, the temperature change reaches 63.2% of its final value at a conductor temperature of $100\ °\text{C} + (119.5 - 100) \times 0.632 = 112.3\ °\text{C}$. In Figure D.1, this corresponds to a time of 11.3 min.

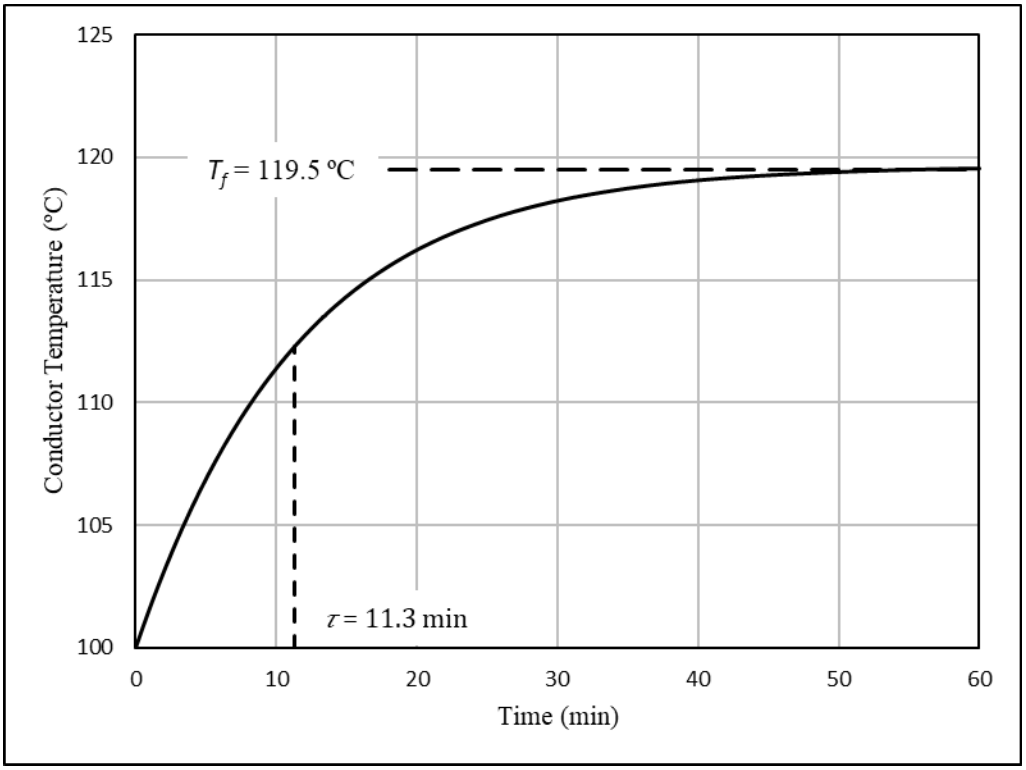

Figure D.1—Conductor temperature vs. time curve for a step change from 1025 A to 1200 A for Drake/ACSR per the assumptions in 4.6.3

Annex E

(informative)

Selection of weather conditions in accordance with CIGRE technical brochure 299

In CIGRE Technical Brochure 299 [B8], the issue of selecting weather conditions for line ratings is discussed at some length based on an exhaustive review of the technical literature. The recommendations provided in [B8] represent a practical guide for developing thermal rating estimates for overhead lines, assuming that the engineer recognizes the need for normal clearance and safety margins employed in the design and operation of overhead transmission lines. There were also a series of qualifying objectives set forth by the brochure for the determination of the weather parameters. The use of [B8] is not required by this standard and is provided here only on an informative basis.

While [B8] makes a series of recommended values, suitable engineering judgment shall be used to determine applicability to the user's unique situation. In particular, for operation above 100 °C certain weather and conductor parameters are increased in sensitivity (for instance, emissivity and wind speed become dominant at high temperatures while solar heat gain becomes relatively minor). [B8] contains a detailed discussion of the sensitivity of these variables that is useful when determining what values to use.

The default recommendations for base ratings in [B8] are based on the likelihood of coincident worst-case rating conditions. For example, while effective wind speeds are sometimes lower than 0.6 m/s, it is extremely unlikely that they are lower than that value if the ambient temperature and solar radiation are also high. Nevertheless, if, for example, ambient temperature and solar radiation are lower, wind speeds lower than 0.6 m/s are more likely.

In IEEE Std 738, indirect and reflected solar radiation are ignored but the direct solar heat intensity can be calculated as a function of date, time-of-day, and latitude. In [B8] it was decided to simply use a solar heat intensity of 1000 W/m^2 direct radiation as a part of most severe coincident conditions. Values of total solar radiation (the sum of direct, indirect, and reflected radiation) can be higher or lower than 1000 W/m^2 varying with time-of-day and season and reflected radiation (caused by ground albedo) can be negligible or as large as 15% to 25% of direct radiation in the visual and 25% to 35% in the near-infrared range.

In short summary, selection of ratings can be chosen using four different levels: base ratings; study-based ratings; ambient-adjusted ratings; or real time ratings.

E.1 Base ratings

Base ratings may be applied for any transmission line and should be used unless the utility adopts alternative practices as described below.

- For sag-limited lines, [B8] recommends that base ratings be calculated for an effective wind speed of 0.6 m/s (2 ft/s), an ambient temperature close to the annual maximum of ambient temperature along the line route and a solar radiation of 1000 W/m^2 (92.9 W/ft^2).
- For those lines where annealing of conductors is the primary concern, having narrow, sheltered corridors, with energized conductors either below tree canopy height or between buildings, the base rating should be estimated based on either a 0.4 m/s (1.3 ft/s) effective wind speed or by reducing the maximum conductor design temperature by 10 °C. Although the average conductor temperature, which determines the line sag, is not likely to be higher than that based on 0.6 m/s (2 ft/s) wind speed, the local effective wind speed in sheltered locations may be significantly lower.

— Seasonal ratings should be based on an ambient temperature close to the maximum value of the season along the line and other criteria above, although the precautions discussed in Section 4 of [B8] should be exercised.

E.2 Study-based ratings

The transmission line owner/operator may base the rating assumptions of selected lines or regions on actual weather or rating studies, provided that:

a) Rating weather studies are conducted in the actual transmission line environment, using the methods recommended in Section 5 of [B8]. If seasonal ratings are applied, such studies include their respective seasons.

b) Alternatively, rating studies can be conducted with devices that monitor line tension, sag, clearance, or conductor temperature. The methods are specified in Section 5 of [B8].

E.3 Ambient-adjusted ratings

Ratings can be adjusted based on varying ambient temperatures measures at the time. These are termed continually ambient-adjusted ratings. In this case, unless real time rating systems are used, the wind speed should be based on the assumption of a more conservative effective wind speed than base ratings. The extensive literature review by the [B8] task force clearly indicates that ambient temperature and wind speed are not independent parameters, higher wind speeds being associated with high ambient temperatures.

If the base rating is to be adjusted for daytime conditions, [B8] recommends the following: if the ambient temperature adjustment is less than 8 °C compared to the temperature selected for base rating conditions (for example, if the base ambient temperature is 35 °C and the actual ambient temperature is between 35 °C and 27 °C), the effective wind speed should be selected as no higher than 0.5 m/s (1.64 ft/s). If the temperature adjustment is more than 8 °C, the effective wind speed should be selected as no more than 0.4 m/s (1.3 ft/s). For nighttime ambient-adjusted ratings (between sunset and sunrise when solar radiation is zero), wind speed should be selected as zero (natural convection only), and solar radiation can also be considered nil. Continually ambient-adjusted ratings can provide technically justified ampacity increases for lines that are designed for low maximum conductor temperatures (e.g., below 60 °C to 70 °C). On the other hand, they will generally not provide technically justified benefits for lines designed for 100 °C or higher temperatures and their use is not recommended.

If a study-based line rating is to be adjusted for ambient temperature, it is recommended to reduce the assumed wind speed to account for correlation with ambient temperature. As with ambient adjustment of base ratings, the wind speed at night may be much lower.

E.4 Real-time ratings

Rather than using "worst-case" weather assumptions, the transmission line owner/operator may elect to use real time monitoring equipment for determining the line rating, provided:

a) Monitoring equipment meets the sensitivity, accuracy, and calibration requirements specified in Section 5 of [B8].

b) It has been verified that the lines that are to be monitored meet the design clearance requirements.

c) Monitors are installed in sufficient quantity to provide statistically valid information of the sag or temperature over the entire length of the monitored circuit. See Section 4.5 and Section 5.6 of [B8] for additional guidance.

d) The operator has the capability of adjusting the line current to the level of standard or enhanced ratings in emergency conditions.

Annex F

(informative)

Steady-state thermal rating for a covered overhead conductor

This standard is explicitly for calculating the current-temperature relationship of a bare overhead conductor; however, it can be adapted to calculate the steady-state ampacity of a covered overhead conductor such as a line wire, a tree wire, or a spacer cable by the inclusion of the thermal resistivity of the covering.

The equations below can be used in conjunction with the steady-state heat balance equations in 4.4 to calculate the steady-state ampacity of a covered conductor. The equations shown are only for a single covered conductor or spacer cable not in contact with other cables and are not applicable to multiplex constructions such as neutral supported secondary cable. The equations are also not applicable to conductors with a surface coating where the thermal resistance is negligible.

Unlike bare conductors, the surface temperature (used to calculate convective and radiative heat loss) is at some temperature lower than the conductor temperature (used to calculate electrical resistance). Therefore, in order to use Equation (2) to calculate ampacity, the convective heat loss, q_c, and radiative heat loss, q_r, shall be calculated at the covering outer surface temperature rather than the conductor temperature. The surface temperature can be calculated by Equation (F.1):

$$T_s = T_c - (q_c + q_r - q_s) \times R_T \tag{F.1}$$

where:

T_s = before covering surface temperature (°C)
T_c = conductor temperature (°C)
q_c = convective heat loss (W/ft or W/m) as calculated by Equation (6)
q_r = radiative heat loss (W/ft or W/m) as calculated by Equation (21) or Equation (22)
q_s = solar heat gain (W/ft or W/m) as calculated by Equation (23)
R_T = thermal resistance (°C-ft/W or °C-m/W) as calculated by Equation (F.2)

The conductor temperature is the temperature at which the ampacity is being calculated at. This is often the conductor maximum operating temperature for the covered conductor or spacer cable as set by the relevant industry standard or as given by the cable manufacturer. For more information on the conductor maximum operating temperature for various conditions see ICEA S-121-733 Section 1.3 [B39].

Radiative heat loss and solar heat gain are dependent on emissivity and solar absorptivity respectively. Coverings will have different emissivity and absorptivity than bare conductor. The emissivity and absorptivity for specific coverings can be obtained from the cable manufacturer. A value of 0.9 for both emissivity and absorptivity is often used for black coverings.

The thermal resistance in Equation (F.1) can be calculated by Equation (F.2):

$$R_T = \frac{R_\lambda \times \ln(D_2/D_1)}{2\pi} \tag{F.2}$$

where:

R_T = thermal resistance of the covering ($°C - \text{ft}/\text{W}$ or $°C - \text{m}/\text{W}$)
R_λ = thermal resistivity of the covering ($°C - \text{ft}/\text{W}$ or $°C - \text{m}/\text{W}$)
D_2 = covered conductor or space cable diameter (same units as conductor diameter)
D_1 = conductor diameter (same units as covered conductor or spacer cable diameter)

The thermal resistivity of the covering can be obtained from the cable manufacturer. Polyethylene is often assumed to have a thermal resistivity of 3.5 °C − m/W (11.5 °C − ft/W).

It can be noted that Equation (F.1) contains both q_c and q_r which are dependent upon T_s. Therefore, the solution to this equation requires an iterative calculation of the surface temperature. This can be accomplished by:

1) Calculating q_c and q_r at some assumed temperature (e.g., T_c).
2) Calculating T_s from q_c and q_r found in step 1.
3) Re-calculating q_c and q_r from T_s from in step 2.
4) Re-calculating T_s from q_c and q_r found in step 3.
5) If the re-calculated T_s in step 4 does not match the value of T_s in step 2 then steps 1 through 4 can be repeated as many times as necessary until T_s no longer changes (i.e., it has reached steady-state).

Once T_s is calculated, the resulting q_c and q_r values at T_s can be used with Equation (2) to calculate the ampacity of a covered conductor or spacer cable.

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